5G for people and things

- Key to the programmable world



5G RAN Architecture

Overview

February 25th, 2016

Outline

Main drivers and use cases

Key concepts

Overall architecture

User- and control-plane architecture

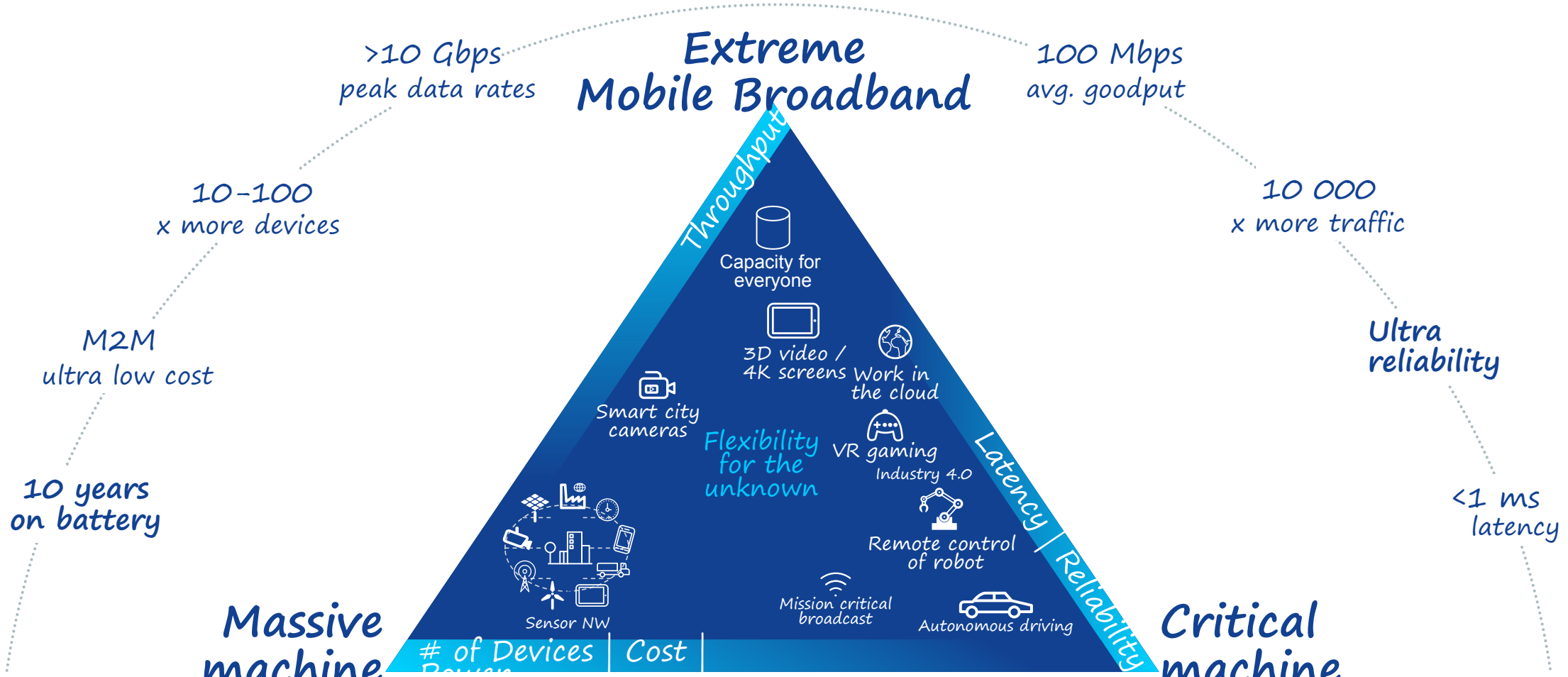
LTE-5G and multi-RAT integration

Multi-connectivity options

Mobility concepts

Main drivers and use cases

5G for people and things – diversity of use cases and requirements



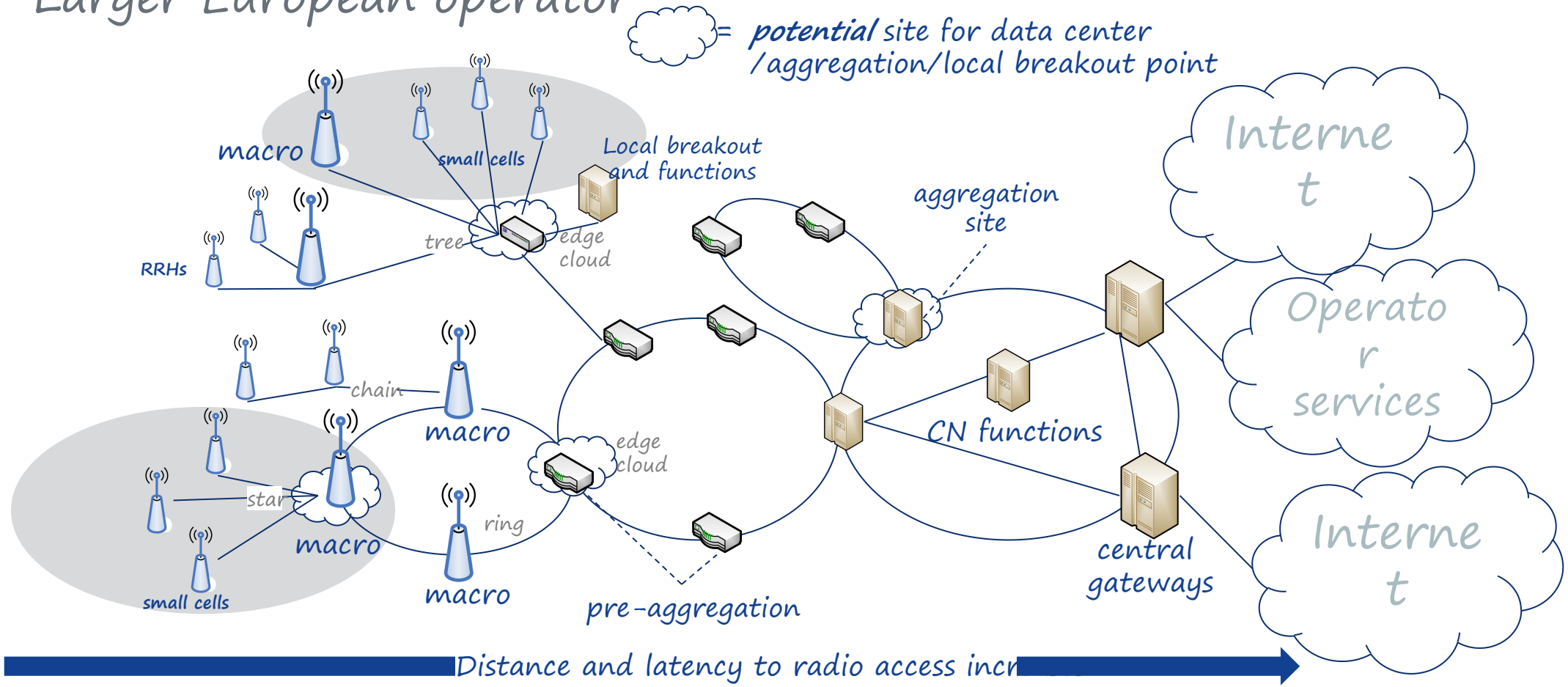
Diversity of Requirements

Several 100 MHz spectrum	HetNet, Ultra Dense Network	Multi-Cell coordination	Reduce latency to milliseconds
New spectrum	High # of Small Cells	At L1 or higher layers	Instant response
Front and backhaul requirements	Coordination among small cells and with macro	how separate entities need to collaborate (eCoMP, dual connectivity, etc.)	Shorter TTI and functionalities "in the box" grow beyond basic eNB functionality (e.g. core functions, …)
distribution	centralization	centralization	distribution

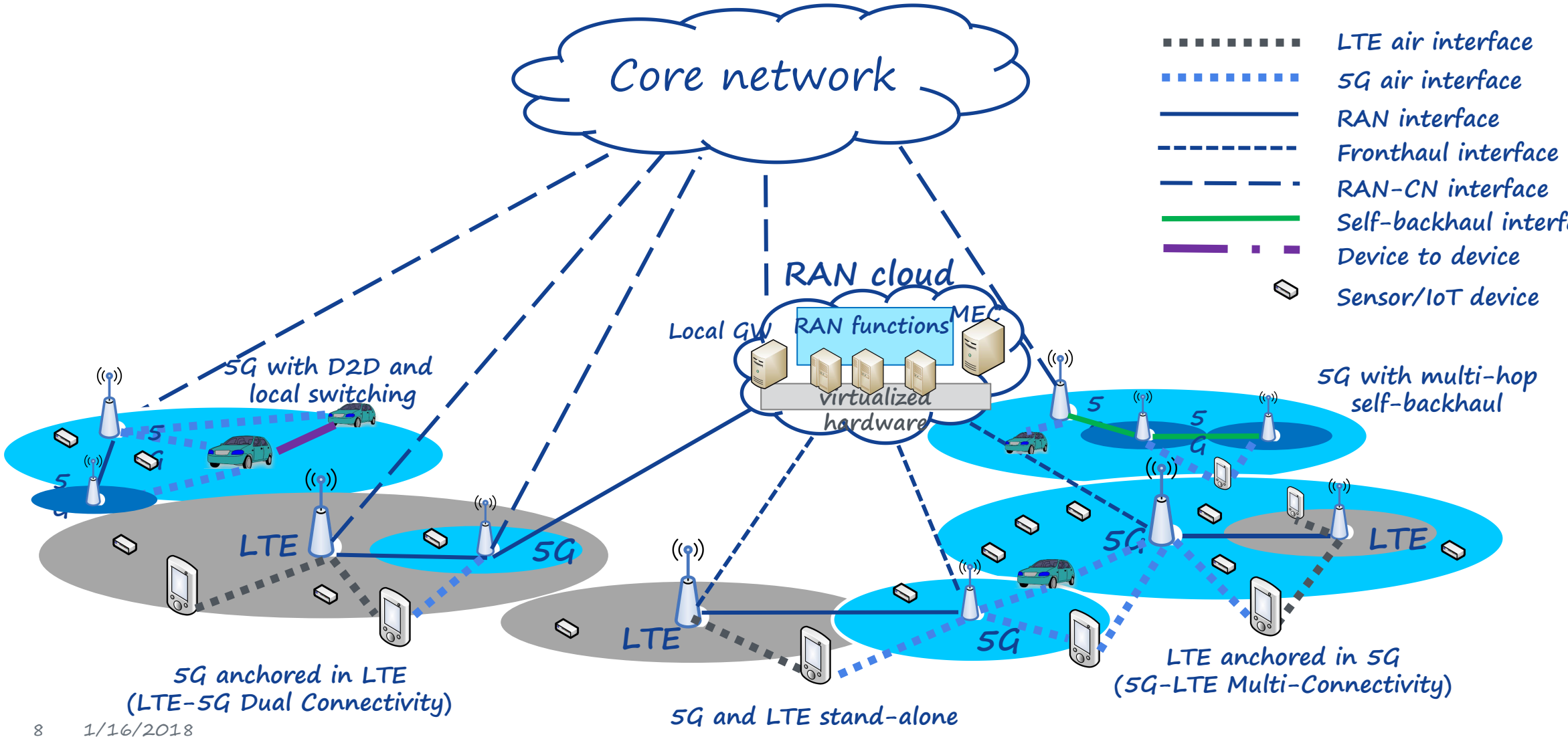
Mobile Network Topology

Larger European operator

➔Diverse x-haul, network topologies, aggregation



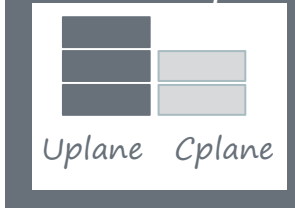
Overall view



Key concepts and objectives

5G Radio Architecture and network design principles

Flexibility and scalability of functions: C/U Plane, Cell/UE related
Load-adaptive scaling



Elasticity

diverse deployment scenarios depending on available xhaul



Any xhaul

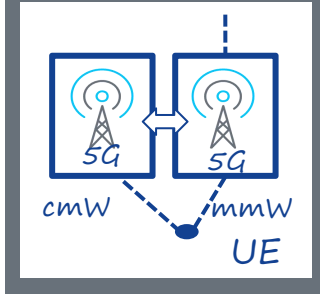
Optimization of control plane and user plane entities

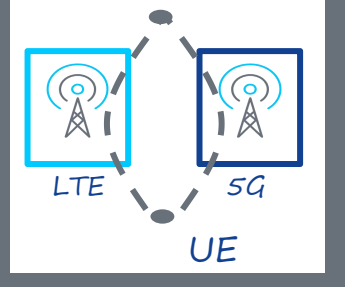
Lean functional architecture

Centralized architecture allows better view to take better decision (e.g. central scheduler, traffic



Integrated Control





Multi connectivity

Objectives

Top-4 Objectives

1) **5G Common Architecture**
   Design cmW, mmW, WA common higher layer architecture and protocol stacks. Considering multi-layer and IRAT: 5G, LTE, WiFi, LAA

2) **Multi-connectivity by design**
   RAN architectural options for spotty 5G coverage and multi-connectivity use cases. Support for central and distributed functions. Support for high throughput and robustness use cases.

3) **Native Cloud RAN**
   Design interfaces and protocol stack to allow flexible functional split acc. to Fronthaul and data center capabilities. Support Cloud implementation, scaling, and dynamic resource allocation. Cloud-friendly protocol design.

4) **Network flexibility**
   Support network slicing, and functional allocation acc. to end-to-end services. Consider SDN approaches by design with c-plane

Scenarios

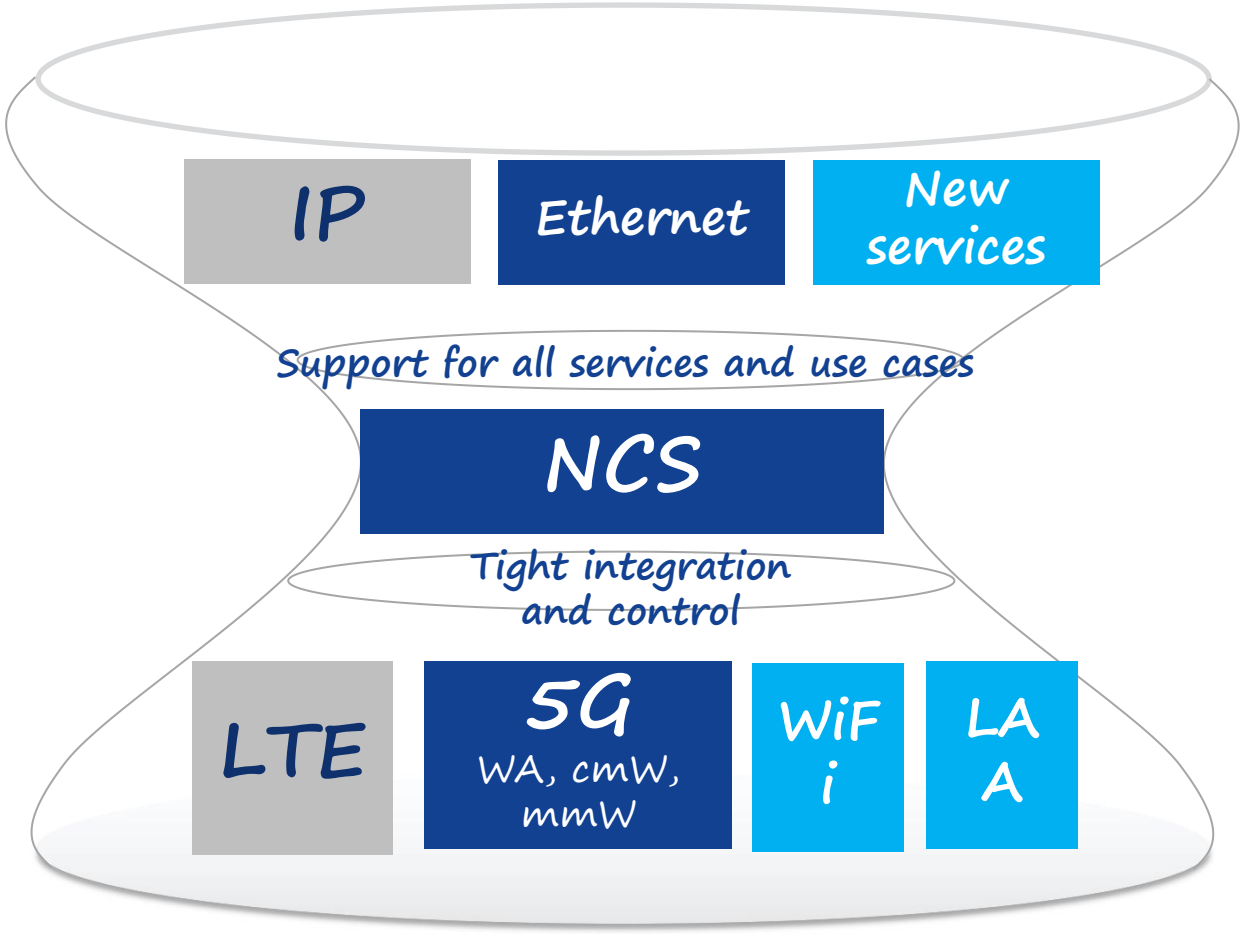
- *5G only: WA, cmW, mmW*
- *Inter-RAT: 5G + LTE + WiFi + LAA*
- *any xhaul*
- *Deployment studies*

Spectrum

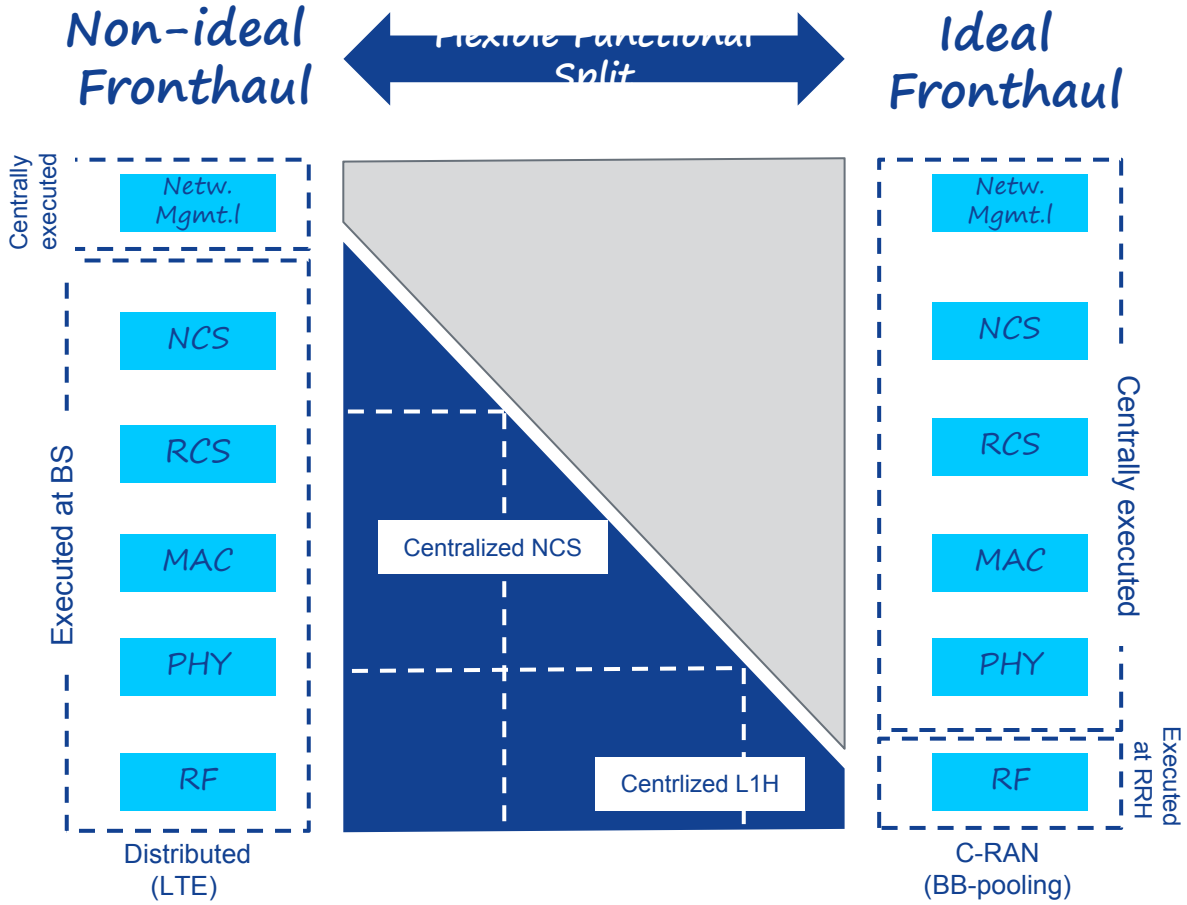
- *Focus is both licensed (and non-shared) and non licensed (WiFi) spectrum cases.*

Key concept: Network Convergence Sublayer

Single layer for all RATs and Multi-Connectivity



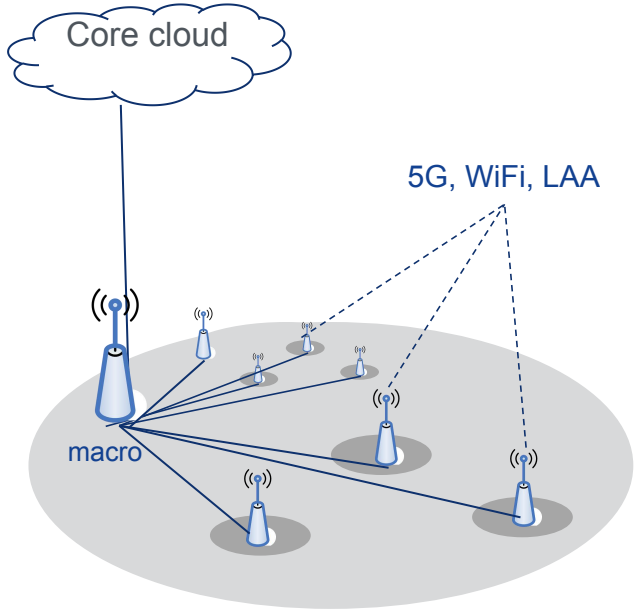
Key concept: Flexible Fronthaul Split
From ideal to non-ideal Fronthaul



Key concept: Topological Flexibility

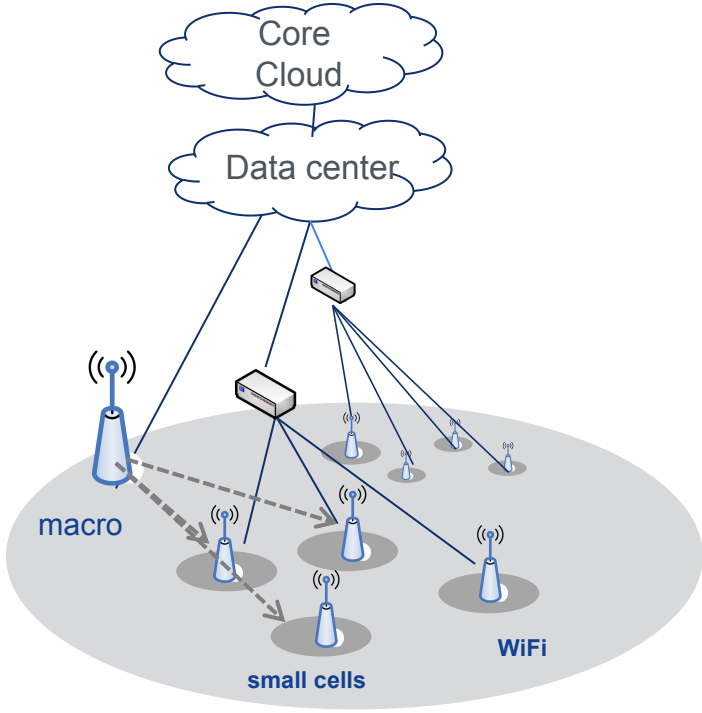
Adapt to operator infrastructure and network topology

1.) Macro-cell as aggregation point



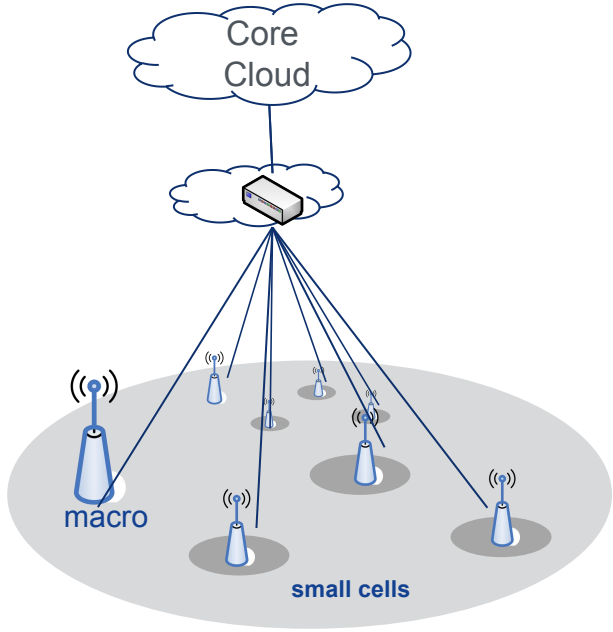
C/U-plane aggregation in the same site (LTE Dual Connectivity baseline)
Issues: U-Plane capacity (processing and fronthaul/backhaul) in macro eNodeB

2.) Dedicated aggregation for small cells



U-plane aggregation nodes for
- reduced x-haul requirements
- distributed processing load

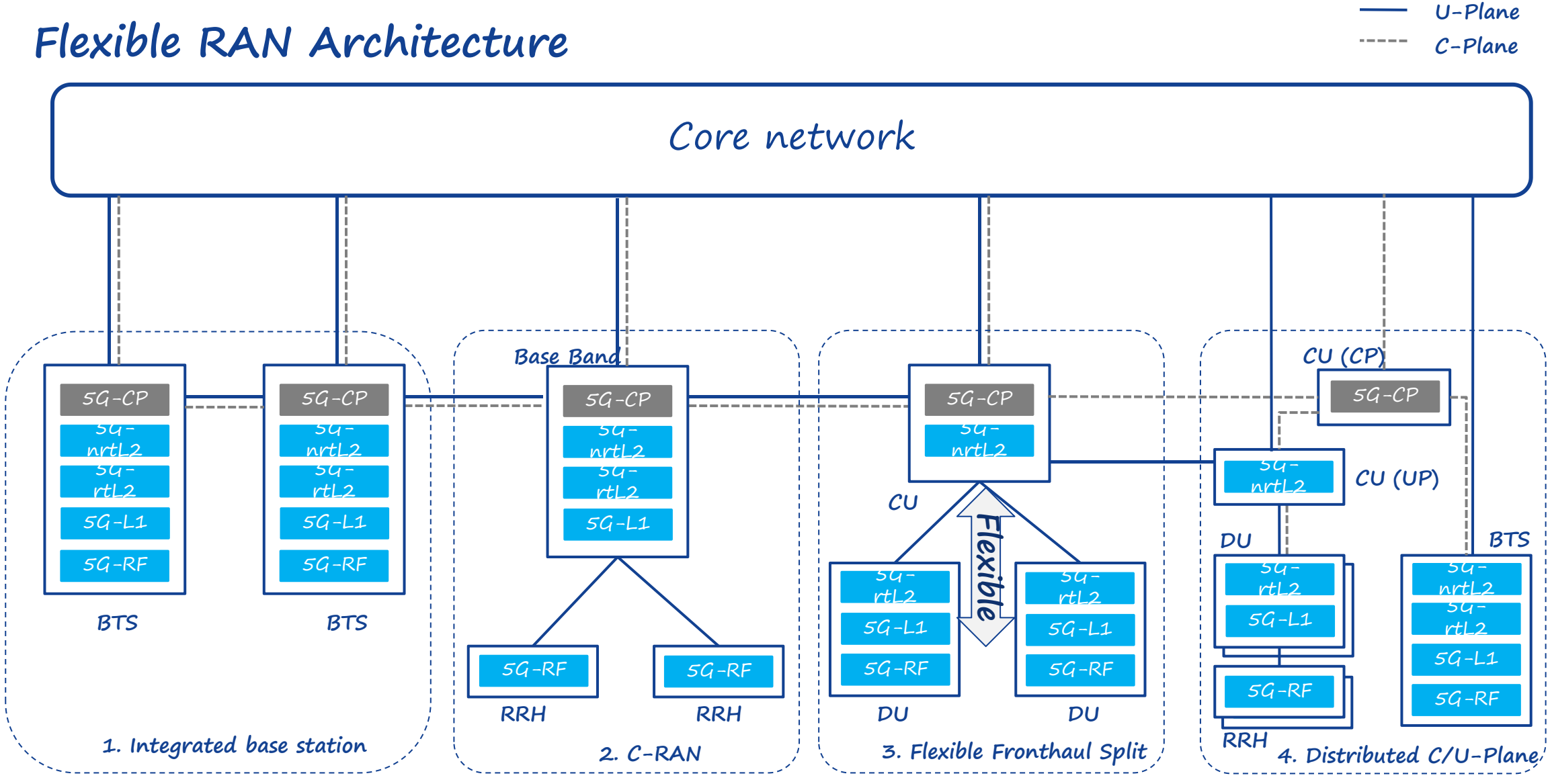
3.) Centralized Scenario



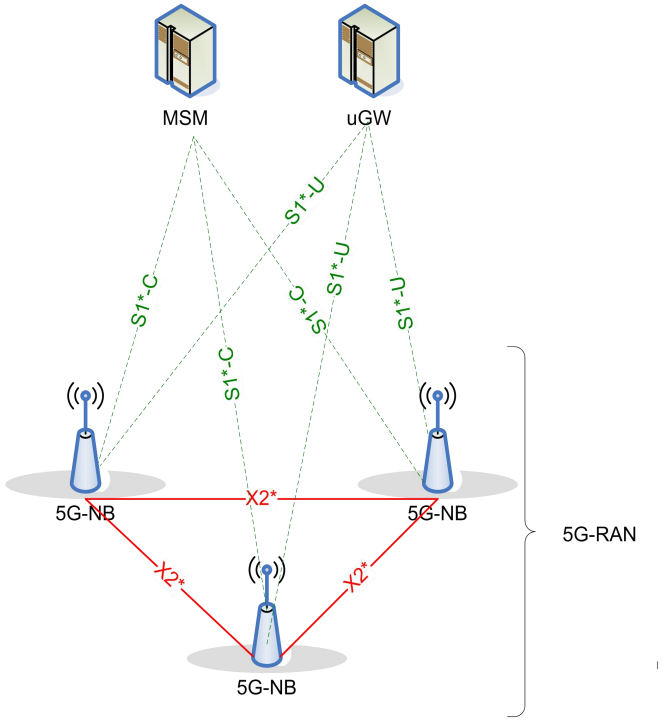
Cloud-RAN scenario with centralized U-plane and C-plane

Overall architecture concept

Flexible RAN Architecture



Basic RAN architecture
Functional split



Basic architecture

MSM: Mobility and Session Management Entity

uGW: User-plane GateWay

5G-NB: 5G-NodeB, radio, 5G-Uu termination

X2*: Interface between 5G-NodeBs

S1*-U: Interface for user-plane data towards uGW

S1*-C: Enhanced control-plane interface

For a detailed functional split description see [1] and [2]

Logical architecture with Fronthaul Split
Distributed 5G-NB configuration

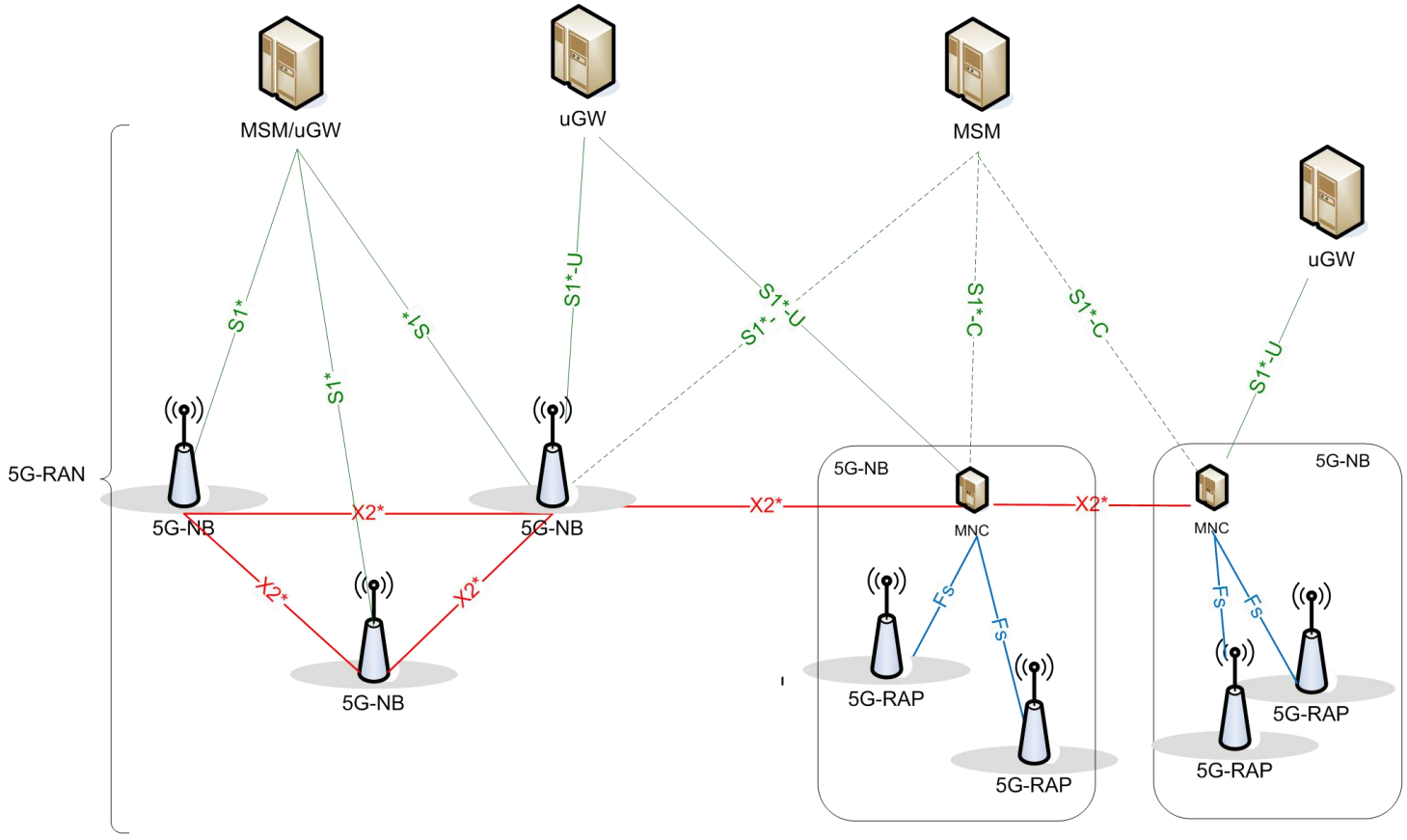
Fronthaul Interface is transparent to NW and UE:
MNC is visible as 5G-NB from NW perspective

MNC manages and controls 5G-RAPs

S1 and X2 interfaces are per MNC, not per 5G-RAPs

5G-RAPs can be macro, small cell, pico cell, mmW, …

NCS in MNC is responsible for multi-connectivity features:
aggregation, steering,



Support for Fronthaul Split and non-co-located User-plane offload

Logical architecture

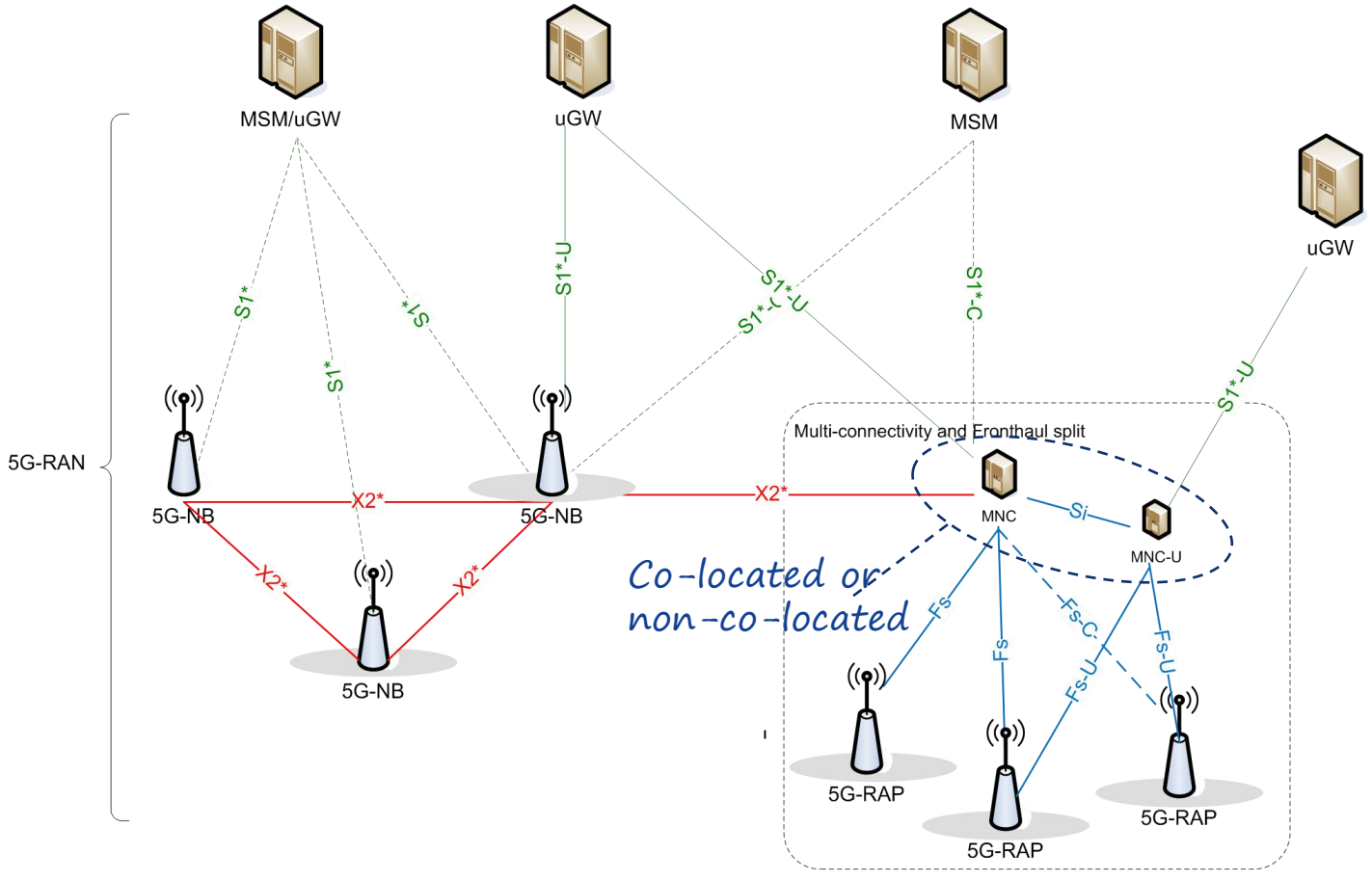
Sidehaul Interface is transparent to network and UE

- Si needs to support for U-Plane configuration and data
- There is a logical 1:many relation between MNC-C and MNC-U

5G-RAP may have interfaces to MNC-C and to one or more MNC-U

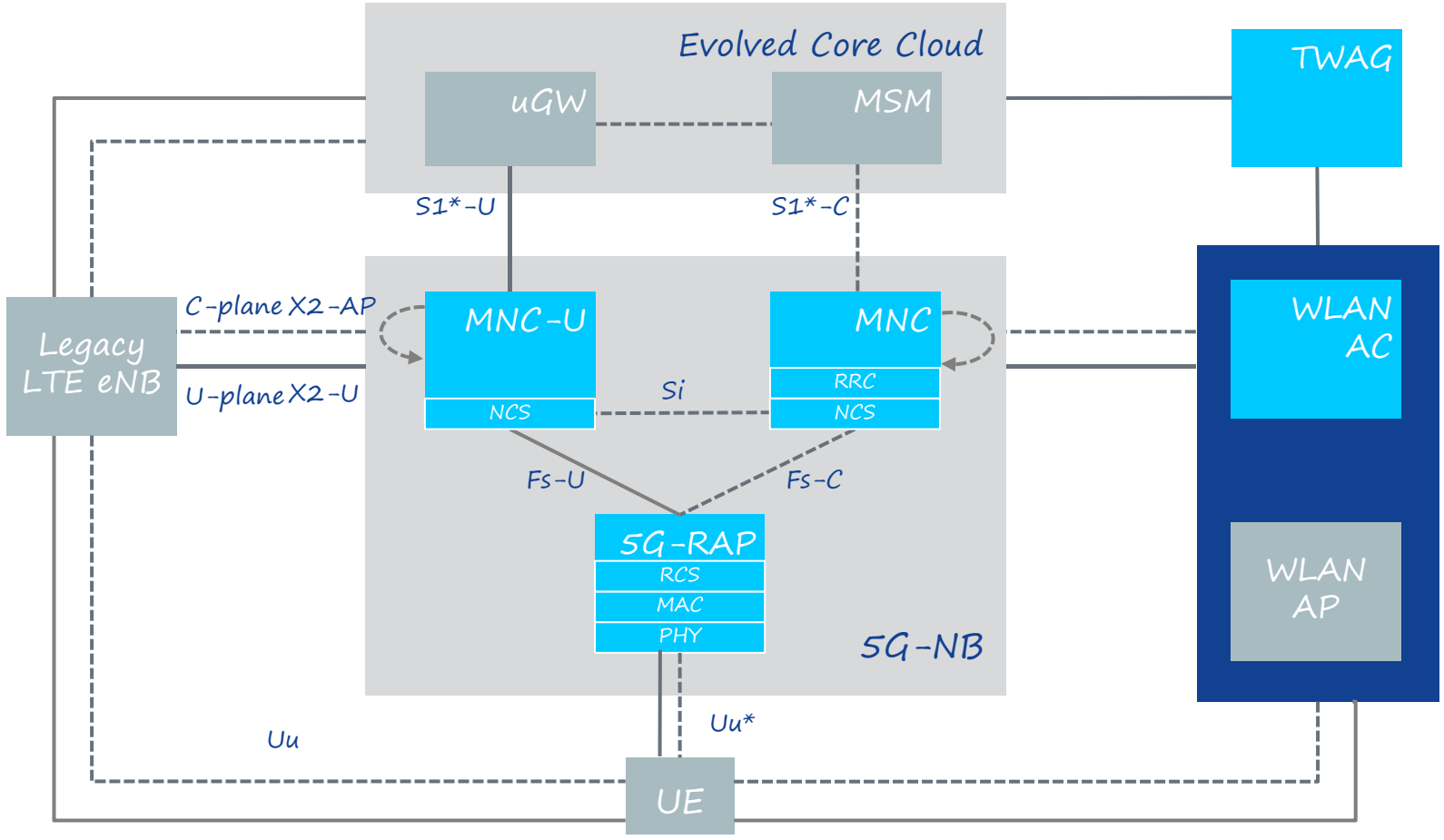
MNC-U terminates S1*-U and X2*-U interfaces towards uGW, 5G-NB

NCS in MNC-U supports split SF to different radio legs



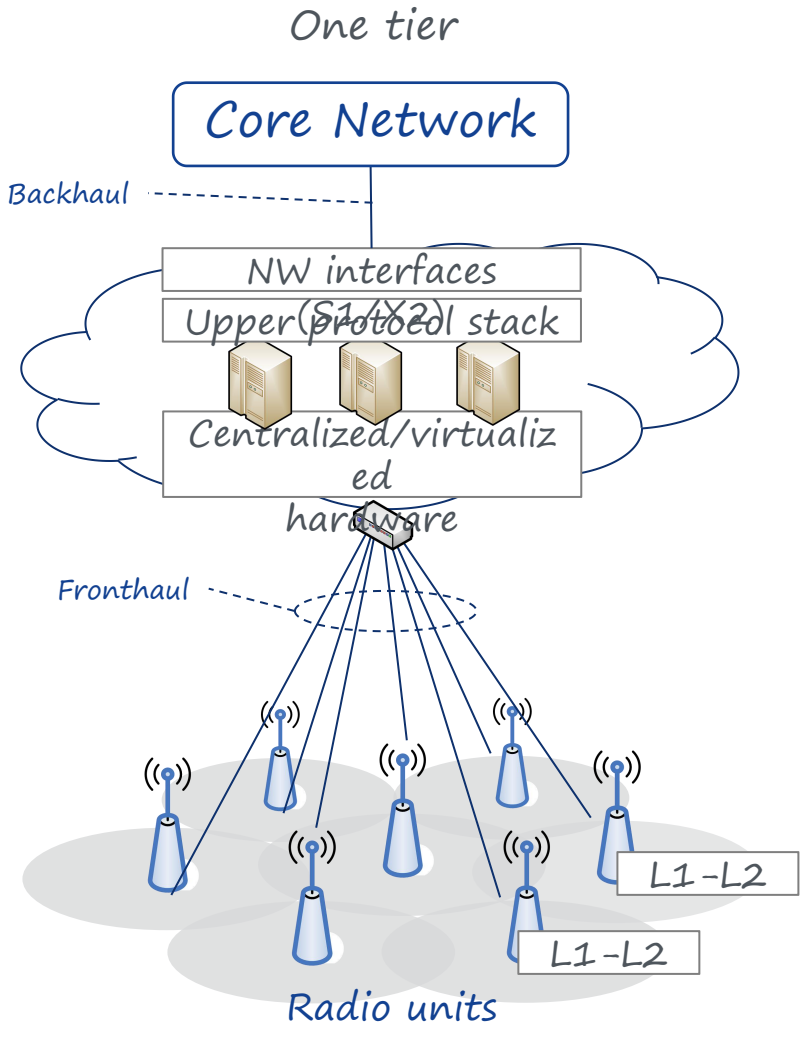
Dedicated node for U-Plane functions

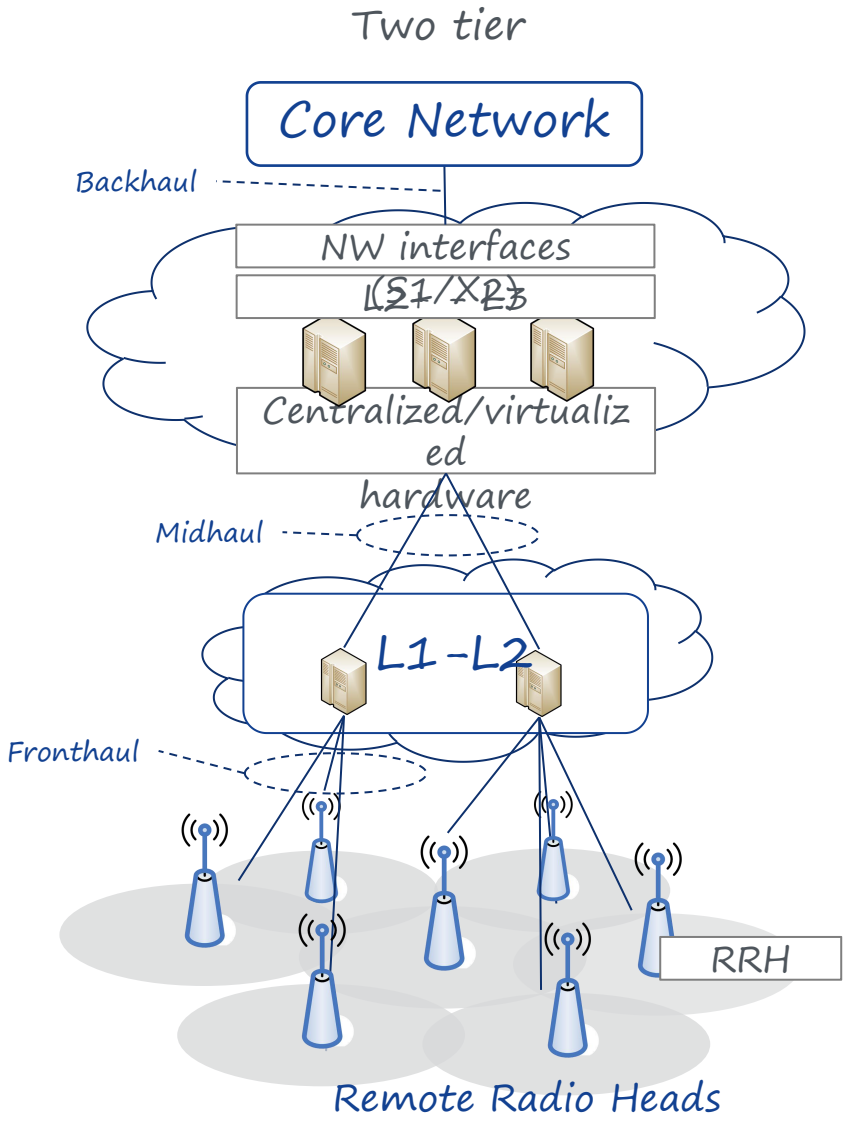
For scaling, offload support



Support for Cloud RAN

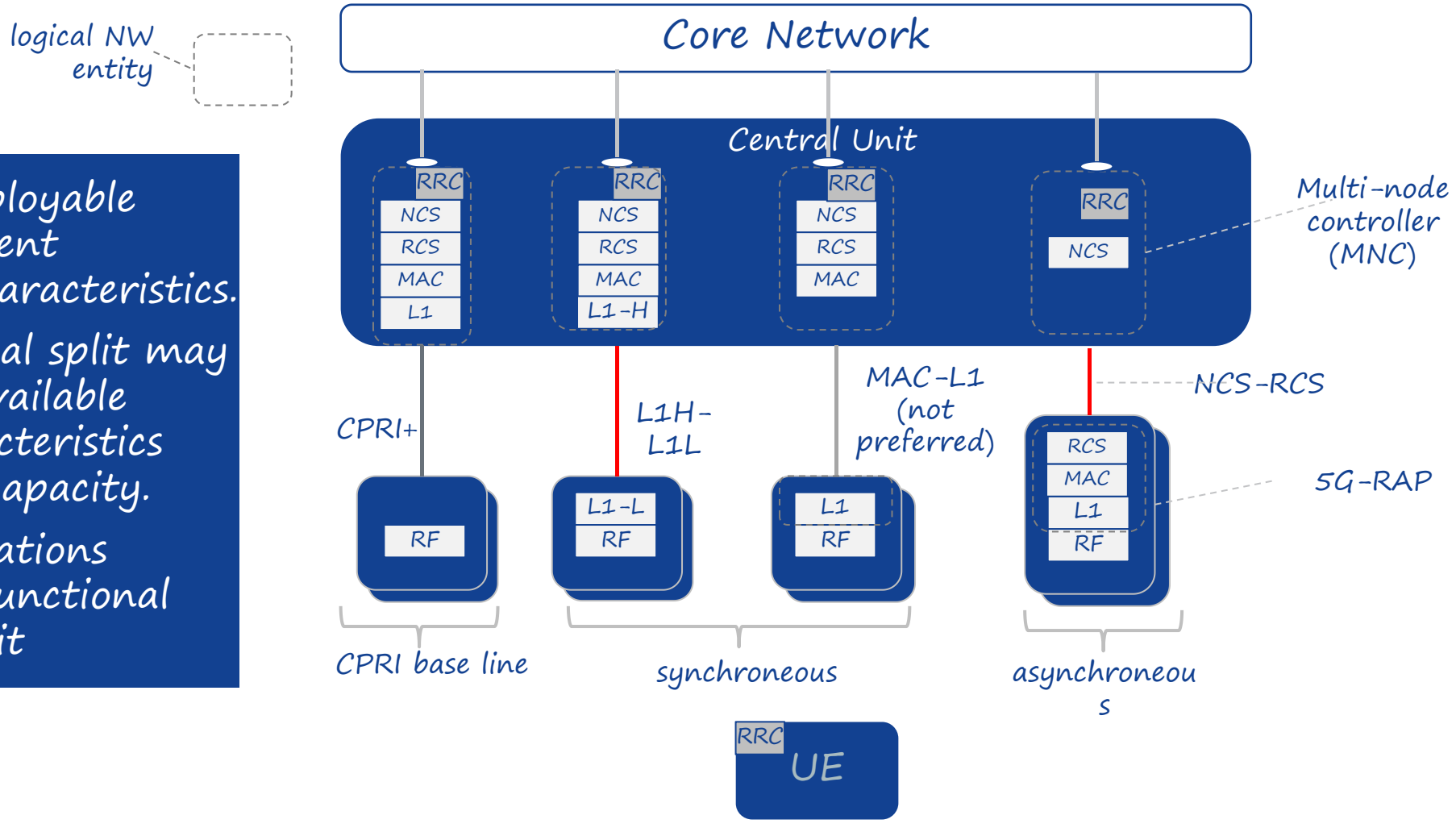
One-tier vs. two-tier Cloud-RAN Scenarios





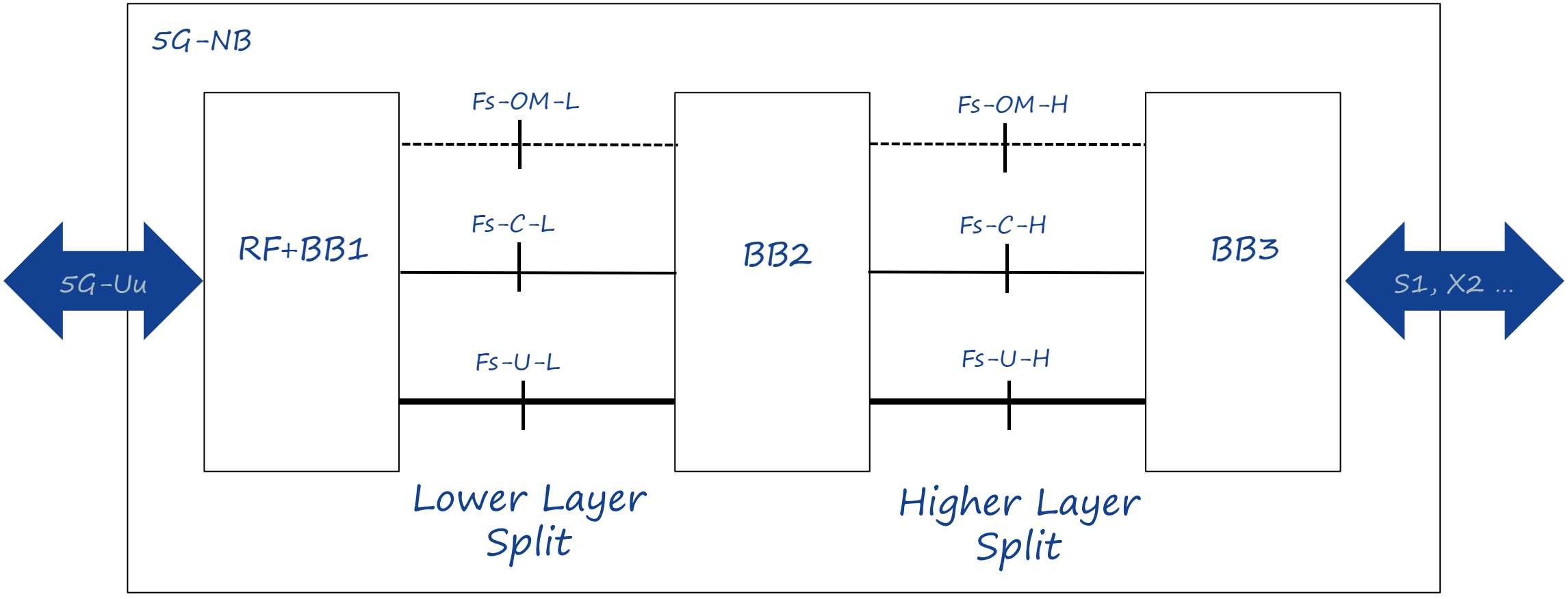
Interface options for Fronthaul Split

- 5G should be deployable supporting different infrastructure characteristics.
- Optimal functional split may depend on the available Fronthaul characteristics for latency and capacity.
- 5G radio specifications should consider functional split and not limit deployments



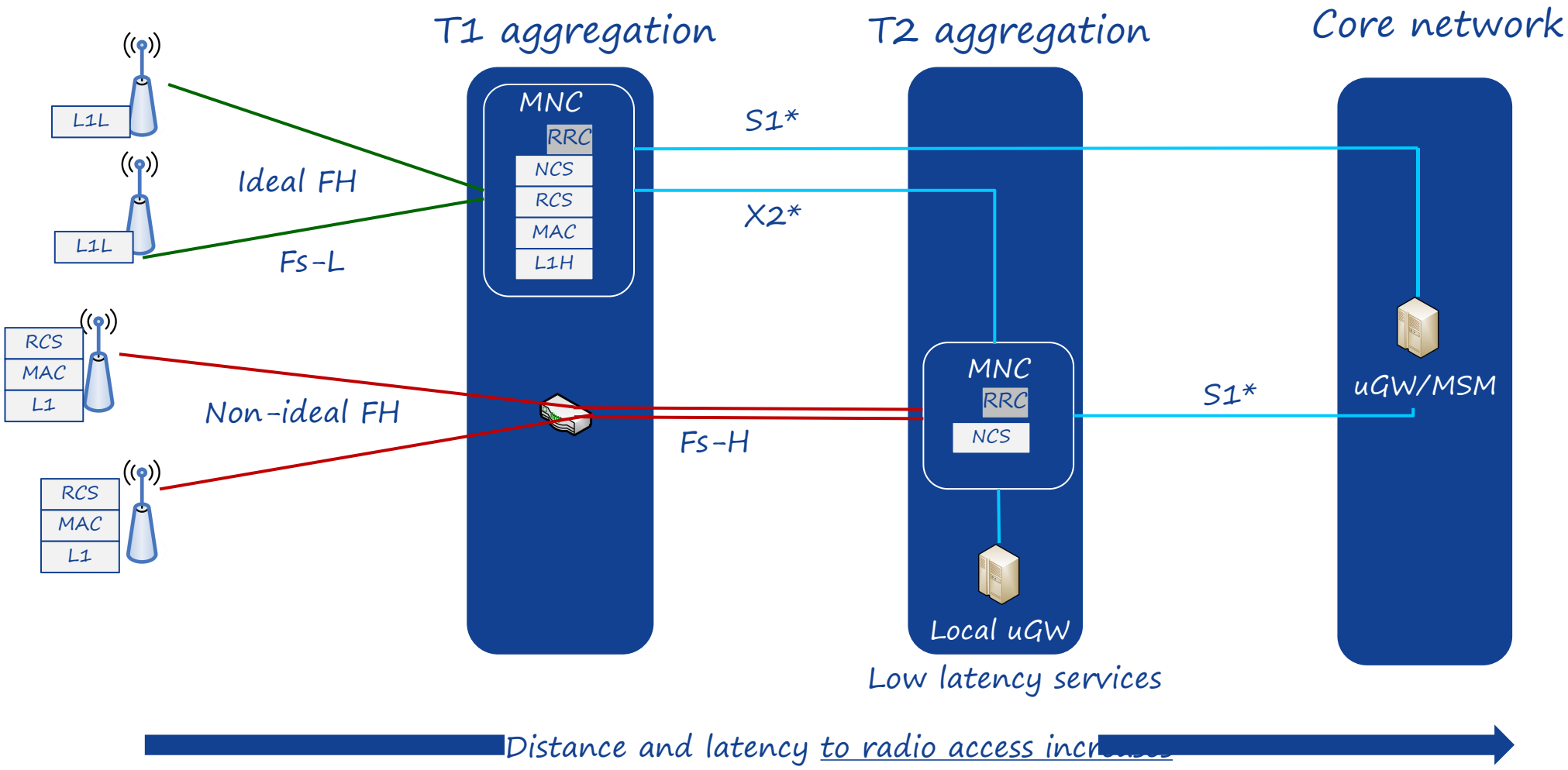
5G Logical Architectural Framework for Fronthaul

Split to Higher Layer and Lower Layer



Support for flexible functional split

Flexibility for Fronthaul and Service Requirements



Multi-Service Support

Multi-service Multi-Connectivity support
Scenario: Inter-5G-NB with Fronthaul Split

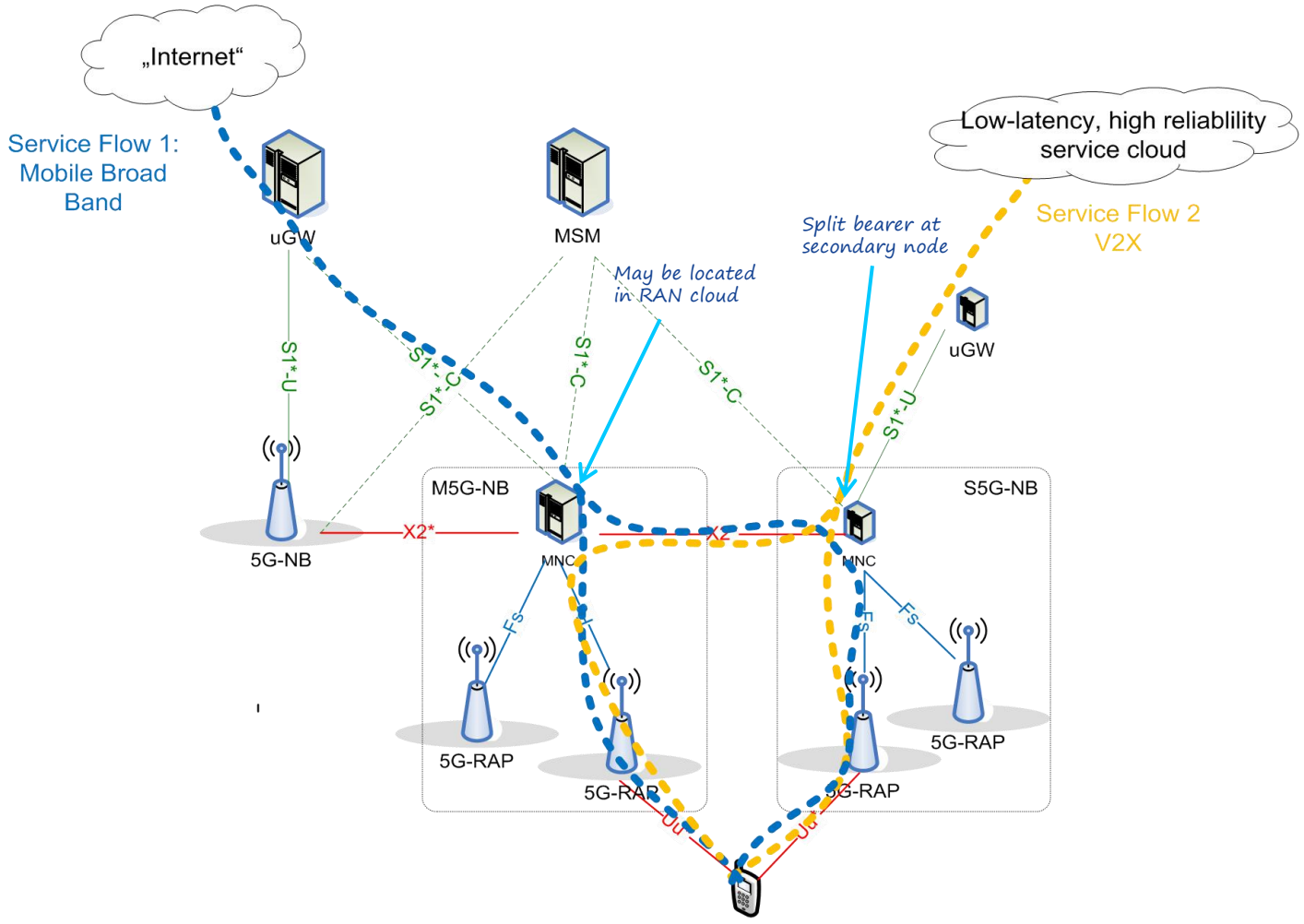
MNC can act as master and secondary node from UE perspective

NCS in MNC responsible for: aggregation, steering, robustness

Multi-connectivity features:

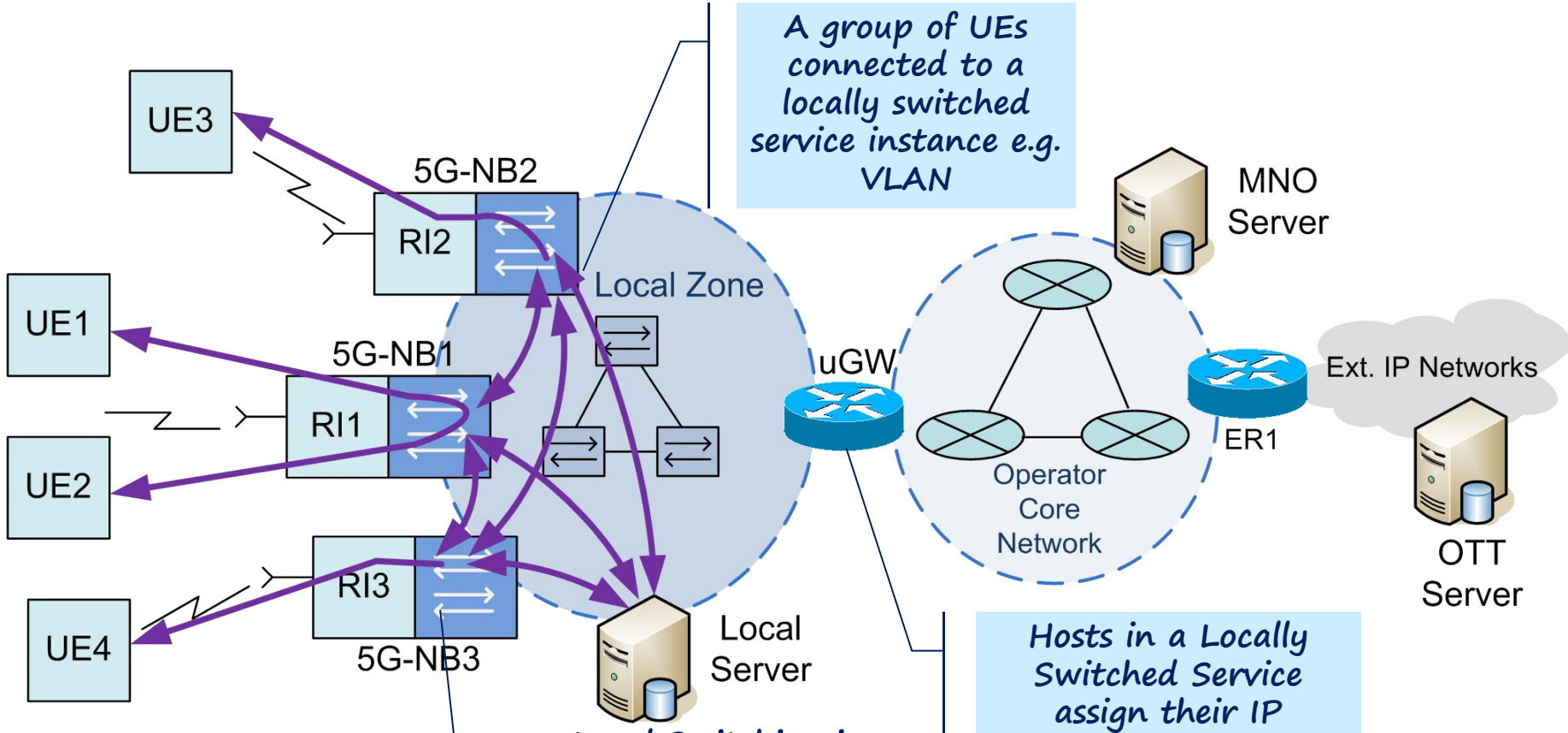
NCS can route to 5G-RAPs over Fs and other 5G-NBs over X2

RRC-M can be dislocated from NCS user plane anchor



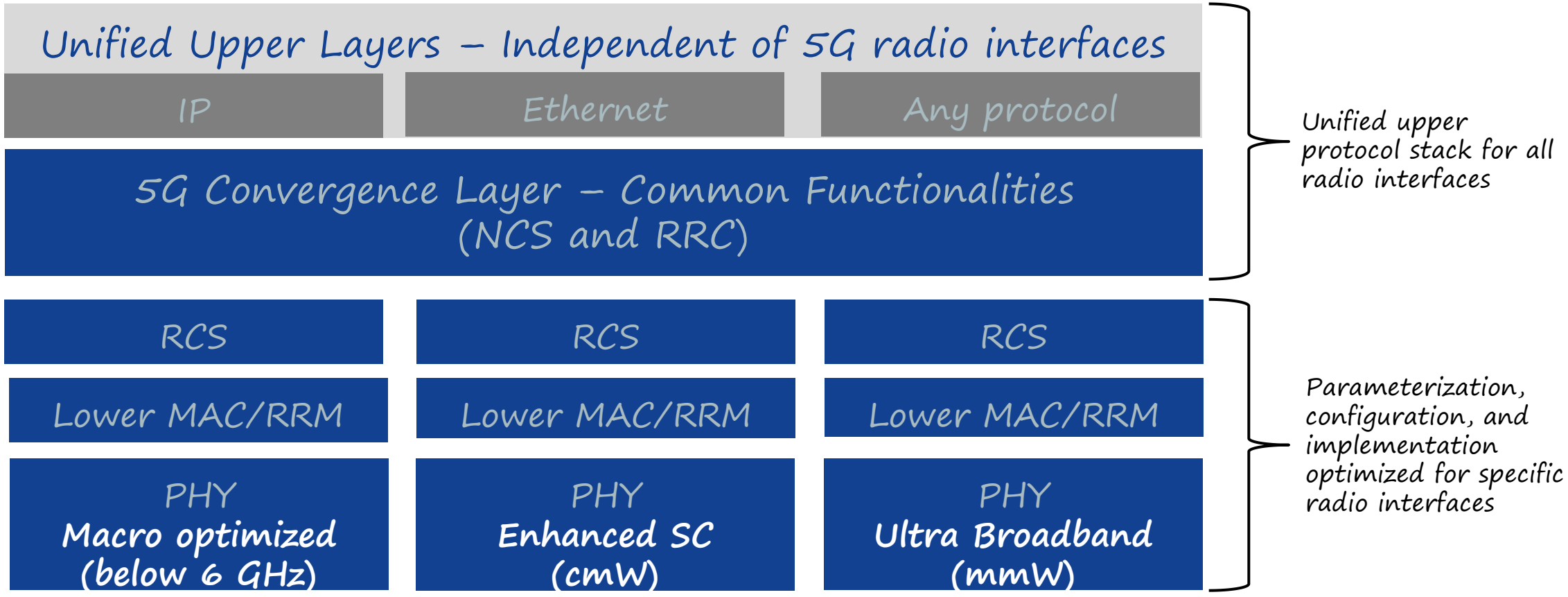
Locally Switched Any-to-Any Connectivity Services below the uGW

5G-NB Modeled with integrated Packet Data Switch



Protocol Architecture

Key concept: Unified protocol stack for 5G air interface



Main Principles

Single common protocol for all radios: NCS (Network convergence sublayer)

- Flexible location, depending on deployment and requirements
- Multi-connectivity traffic handling for U-Plane and C-Plane: routing, splitting, duplication

Common RRC for centralized control of multi-connectivity functions

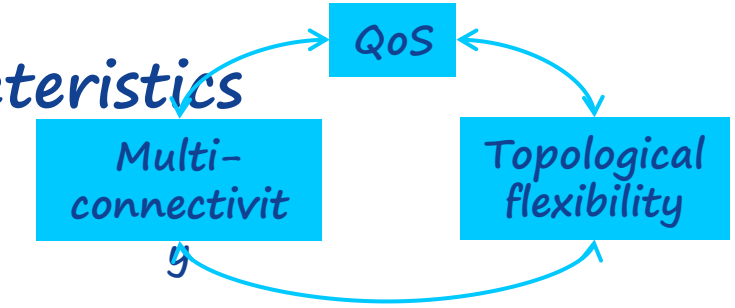
- Reduce signaling and coordination complexity
- Flexibility for different multi-connectivity scenarios
- NCS as transport layer for signaling service flows

Radio Convergence Sublayer (RCS) and Unified MAC

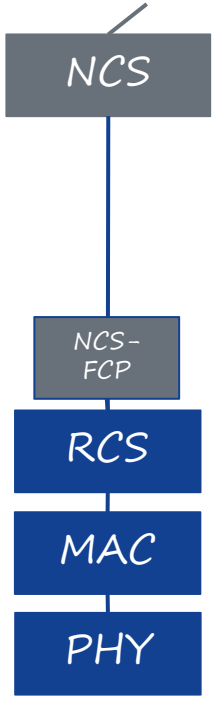
- Parameterizeable for 5G use cases and requirements based
- Optimized for parallel processing in Cloud

Application control scheduling for QoE management and enforcement

NCS Key Functionality and Architecture Characteristics



Service flow



Participate in leg addition/deletion
Partition traffic across multi-connectivity legs

Multi-connectivity anchor

Allows granular handling of UE traffic and multi-connectivity per-SF – e.g. different locations for high- and low-latency SFs

Allow Multiple NCS instances per UE – one per Service Flow

NCS location in network can be based on QoS, network topology, multi-connectivity state, processing load
Network-topology awareness using e.g. SDN

Flexible selection of NCS placement

Awareness of state of lower-layer buffers, path and queueing delays, etc. on each multi-connectivity leg

On DL Flow Control aids partitioning of traffic

On-the-fly, identify sub-flows within SF
At receiver, re-ordering per-sub-flow

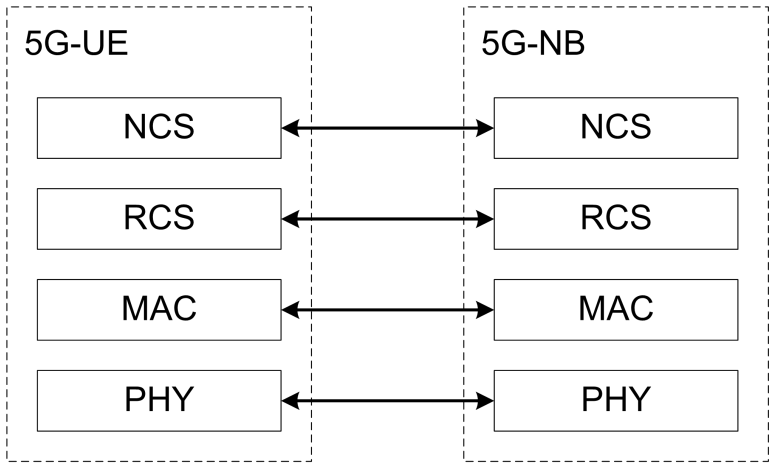
In-Service-Flow Differentiation

Flexible adaptation to change in flow characteristics, multi-connectivity, processing overload

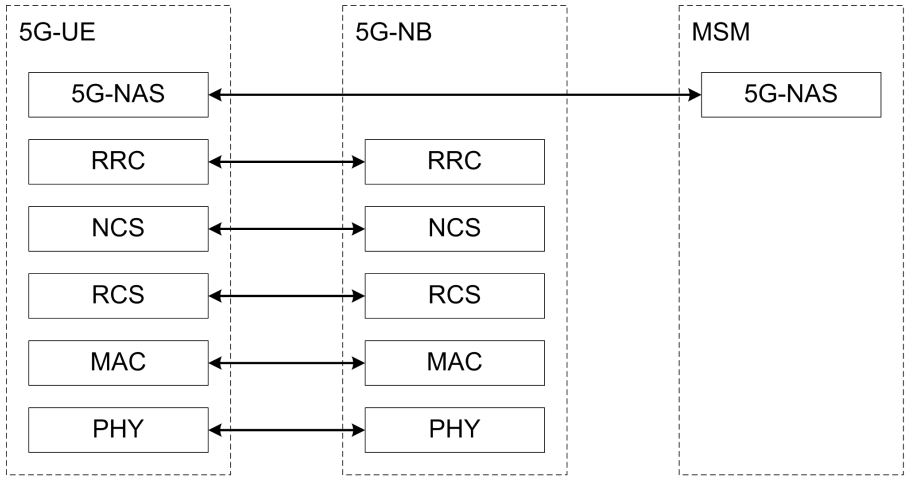
NCS can be relocated

Protocol stack

User plane



Control plane



5G-UE: 5G User Equipment

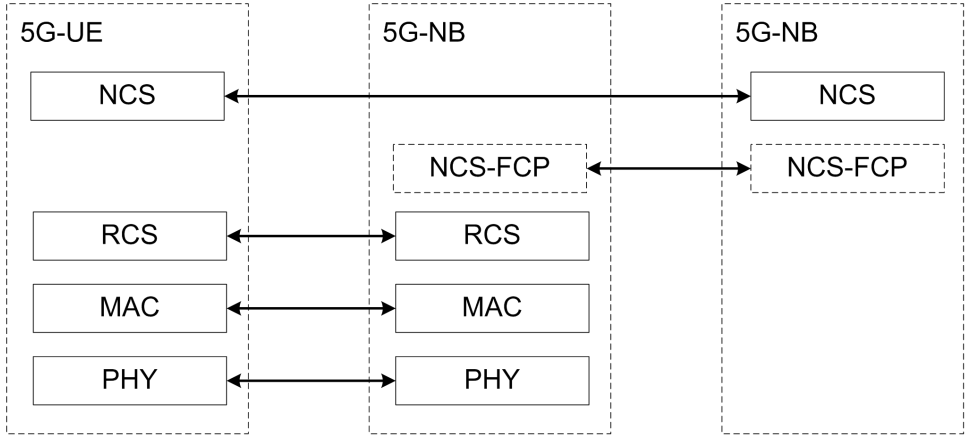
5G-NB: 5G NodeB

NCS: Network Convergence Sublayer

RCS: Radio Control Sublayer

Protocol stack
With support for Multi-Connectivity

User plane



NCS: Network Convergence Sublayer

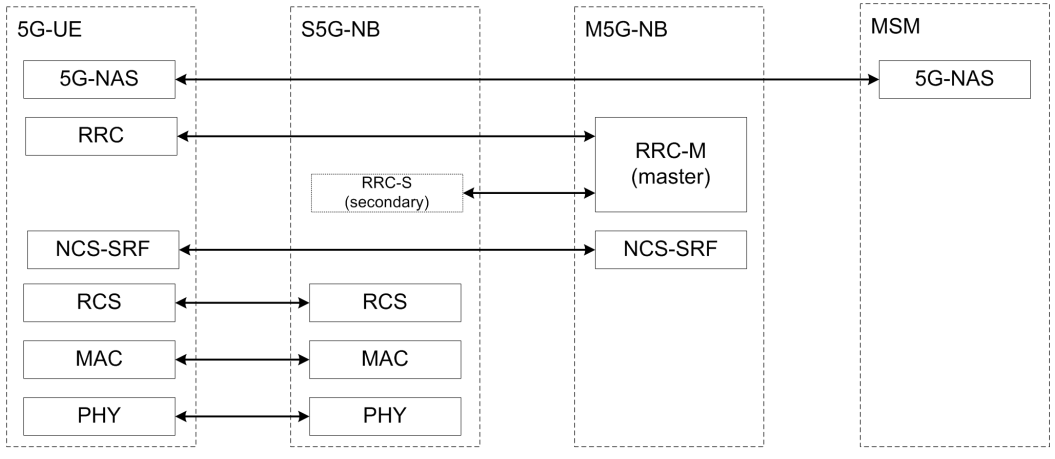
RCS: Radio Control Sublayer

5G-UE: 5G User Equipment

5G-NB: 5G NodeB

NCS-FCP: NCS Flow Protocol (interface AP)

Control plane

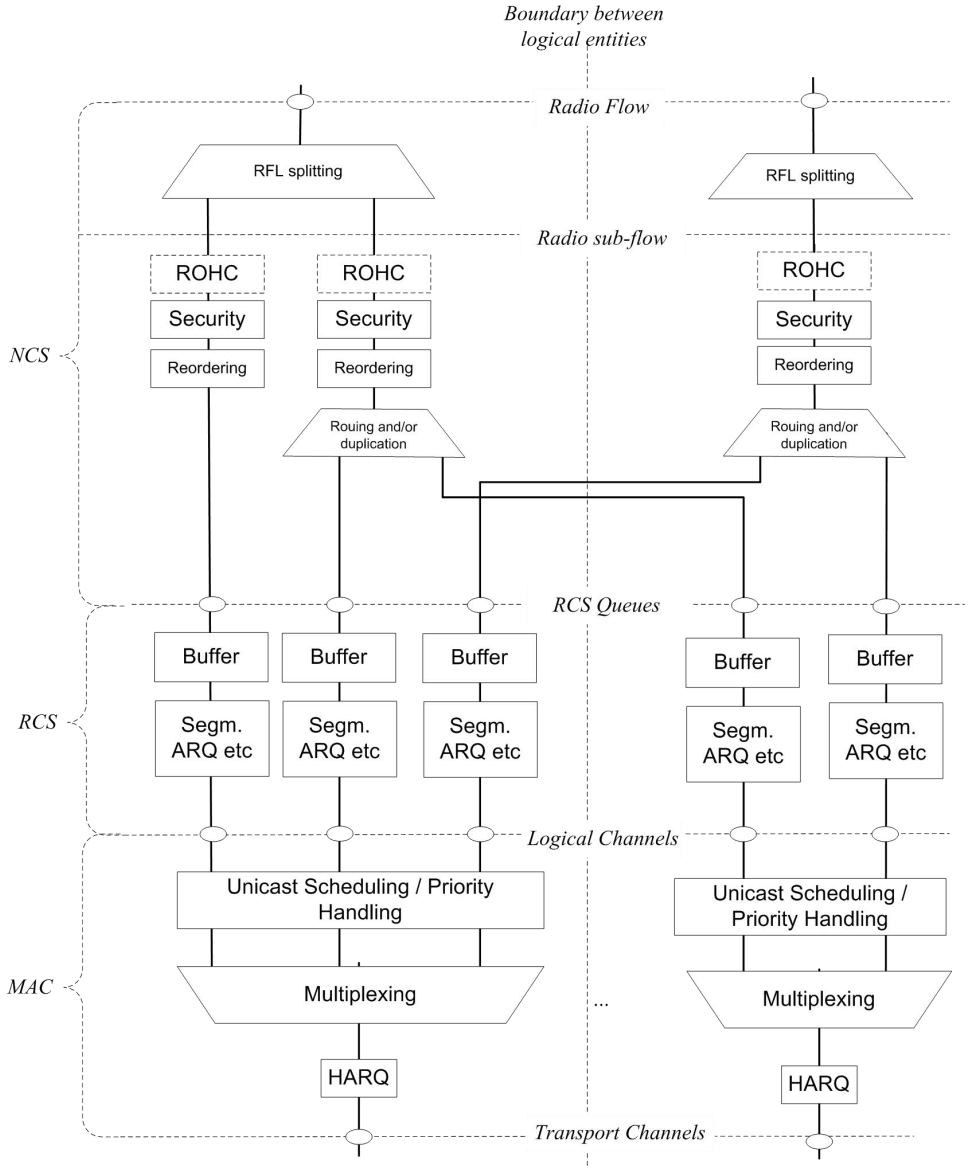


MSM: Mobility and Session Management Entity

uGW: User-plane GateWay

NCS-SRF: NCS instantiated for an SRF

User plane protocol architecture



NCS: Common convergence layer for UP, CP, traffic; Splitting, selection, duplication of packets; selection of radio legs; within the network, may not be co-located with RCS.

Radio Flow: RFL is the service provided by the Layer 2 for transfer of data between User Equipment and 5G-RAN.

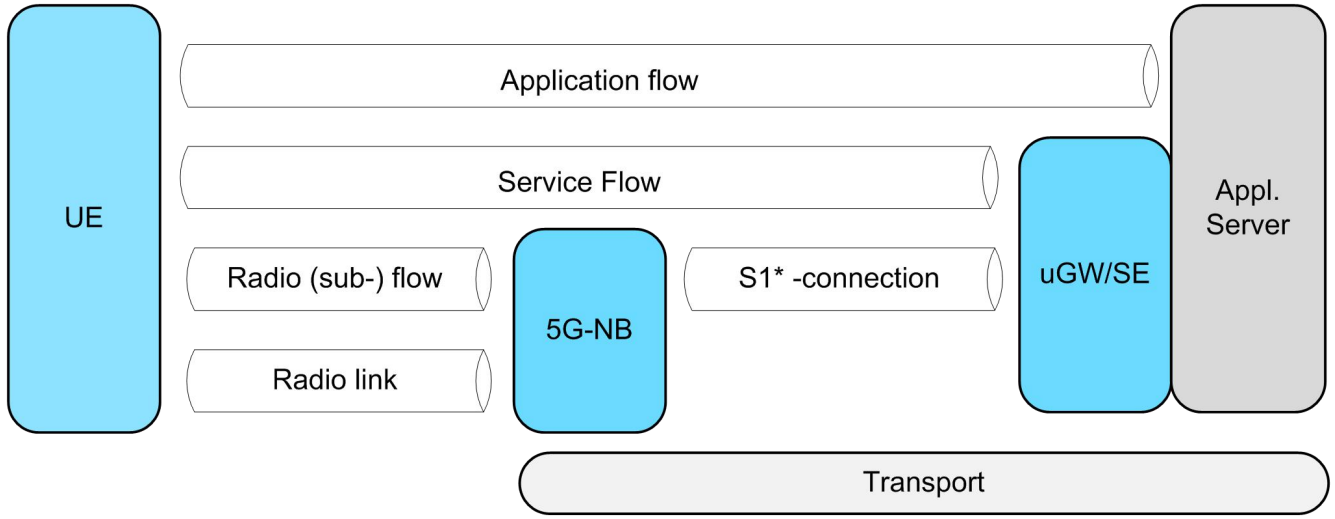
Radio Sub Flow: The service provided by NCS layer for transfer of data between UE and 5G-RAN mapped to a specific QoS Queue.

RCS: Buffering, Segmentation, ARQ, re-ordering

MAC: Multiplexing to transport channels, radio/QoS scheduling

QoS Architecture

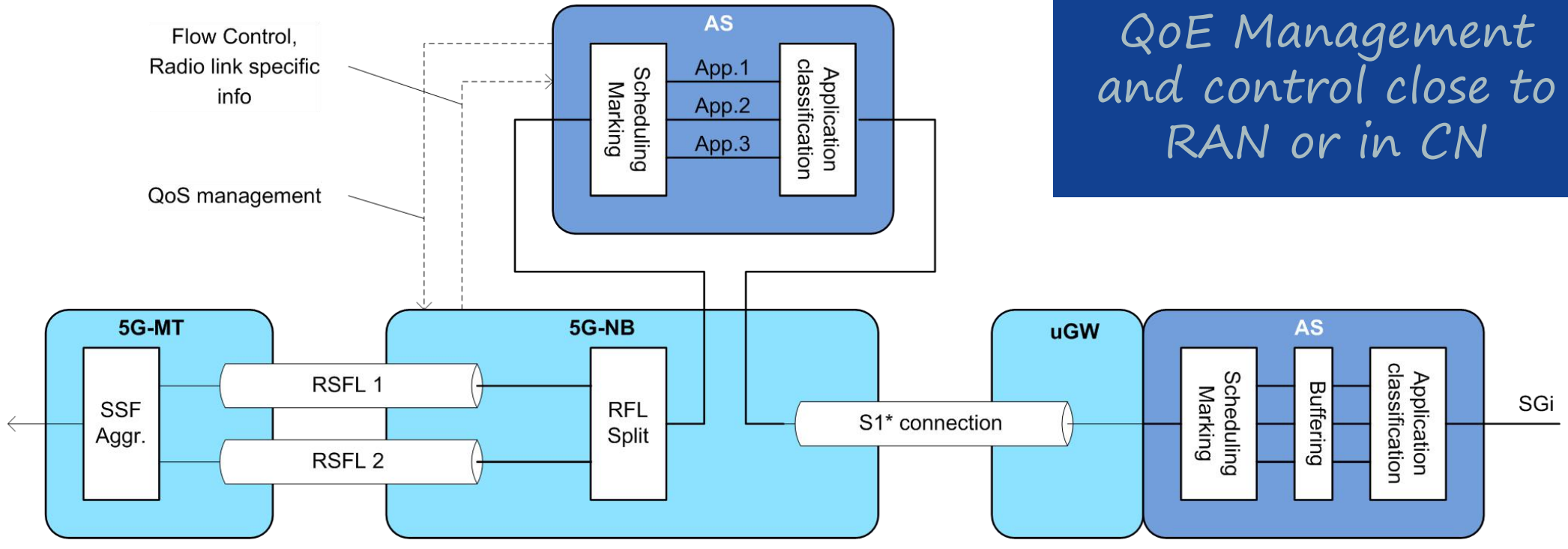
QoS Architecture



- Service flow (SF) provides end to end connectivity between the UE and the Service edge
- S1* connection delivers service flow packets between the RAN and the uGW
- Radio Flow and Radio Sub-Flows (RSFL) transport SF packets between the 5G-NB and the UE. The data packets of the SF can be mapped to multiple RSFLs, each representing a radio level QoS class.
- A radio link transports RSFL packets between the UE and the AP.
  - There is one to one relationship between the RSFL and the radio link in single connectivity case.
  - In the case of multi-connectivity, RSF can be mapped to multiple radio links each belonging to different radio.

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Application scheduler (AS)



QoE Management and control close to RAN or in CN

- AS classifies application flows into sub-service flows (SSFs) and performs scheduling
- The AS informs the 5G-NB about the specific service attributes and the identification of SSFs.
- Based on the received SSF level information, 5G-NB configures its radio stack
- The RFL split function of the AP maps data packets to RSFL based on the SSF information.

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Mapping from NCS to RCS and QoS queues

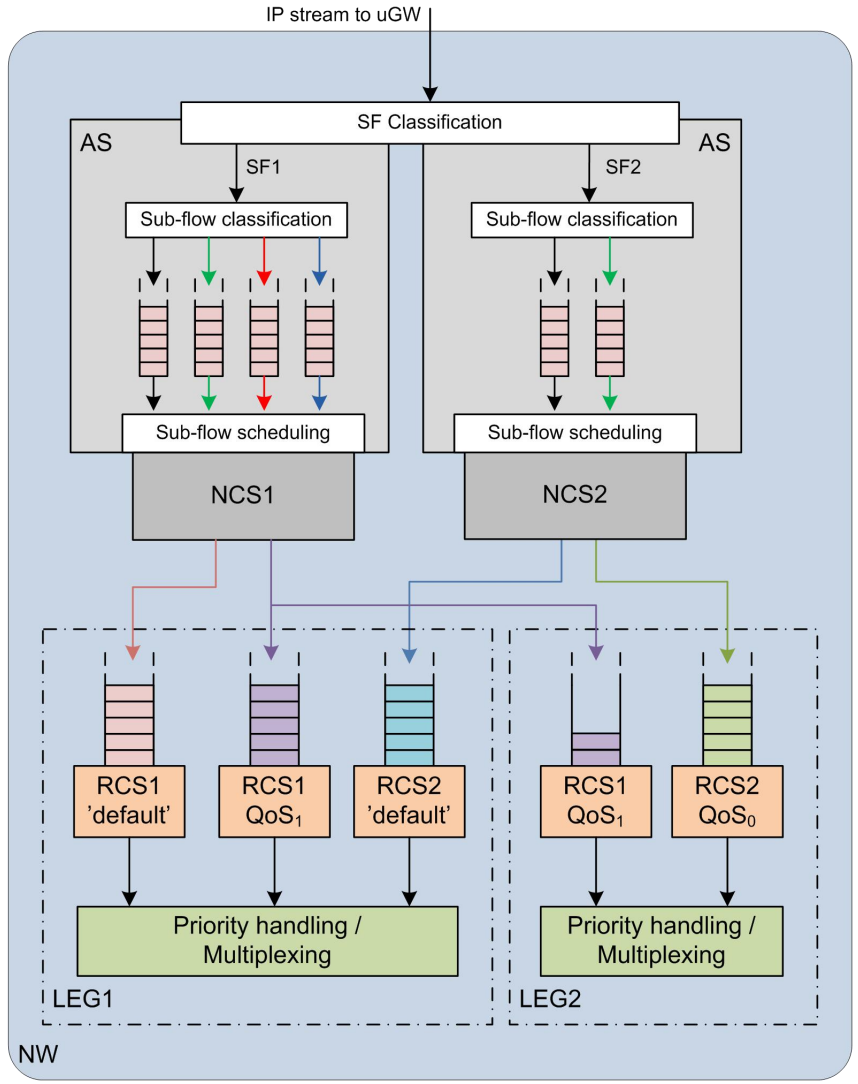
Example of with two radio legs (multi-connectivity) and two service flows

Basic principles:

- One NCS entity serves one Service Flow (radio flow)
- One RCS entity serves one NCS entity
- Multiple RCS entities can serve the same NCS entity

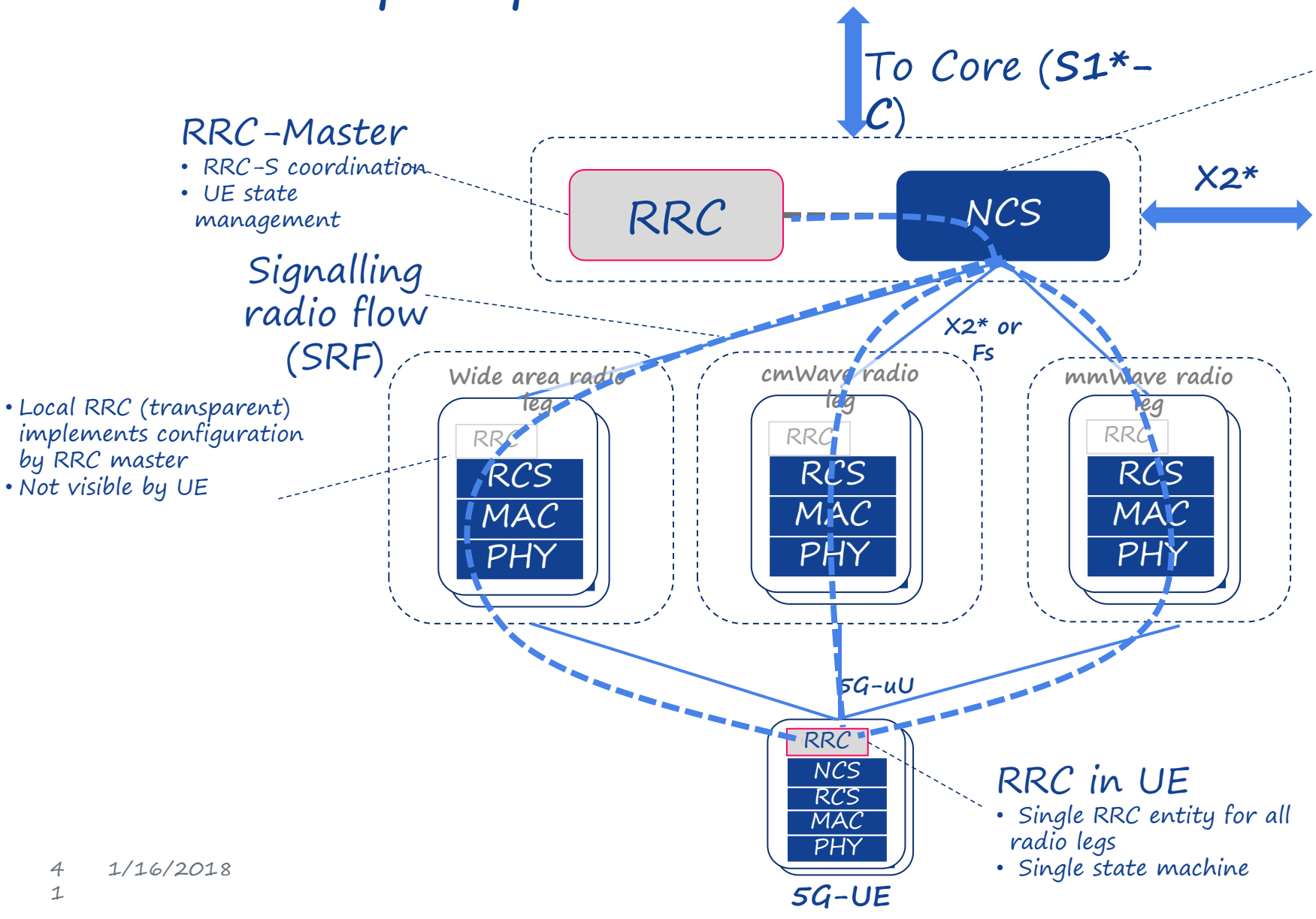
Configuration, setup and operation:

- During setup, at least one RCS entity is configured for the setup NCS entity with default configuration and QoS.
- The setup and termination of radio (sub-)flows is performed by RRC.
- NCS can route packets to multiple RCS entities in different radio legs belonging to the same Radio Service.
- Radio scheduler/MAC has no visibility to application flows or a radio services but sees the



Control Plane - RRC

Control-Plane principles

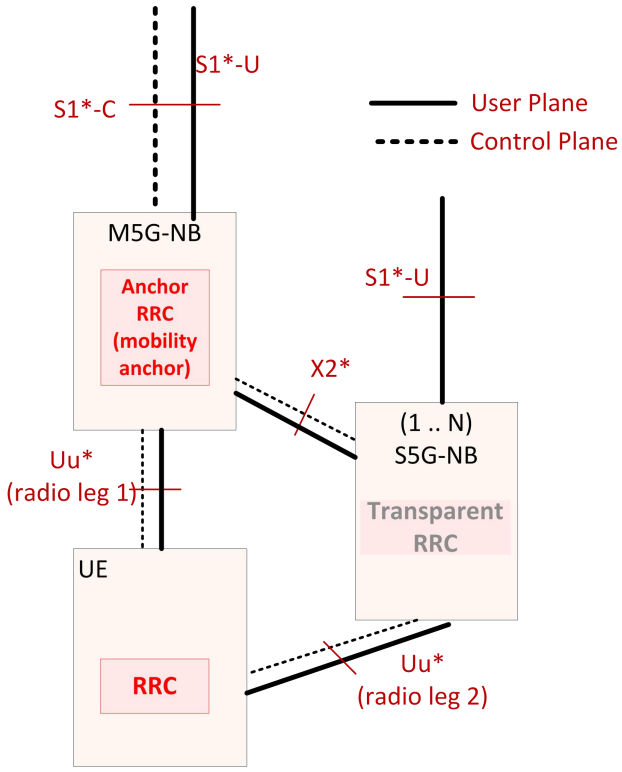


Key concept:
Separation of logical from transport plane for control messages

Logical view:
One RRC entity in UE communicates with RRC-M over different radio legs

Transport view:
NCS performs message routing for RRC from and to different radio legs

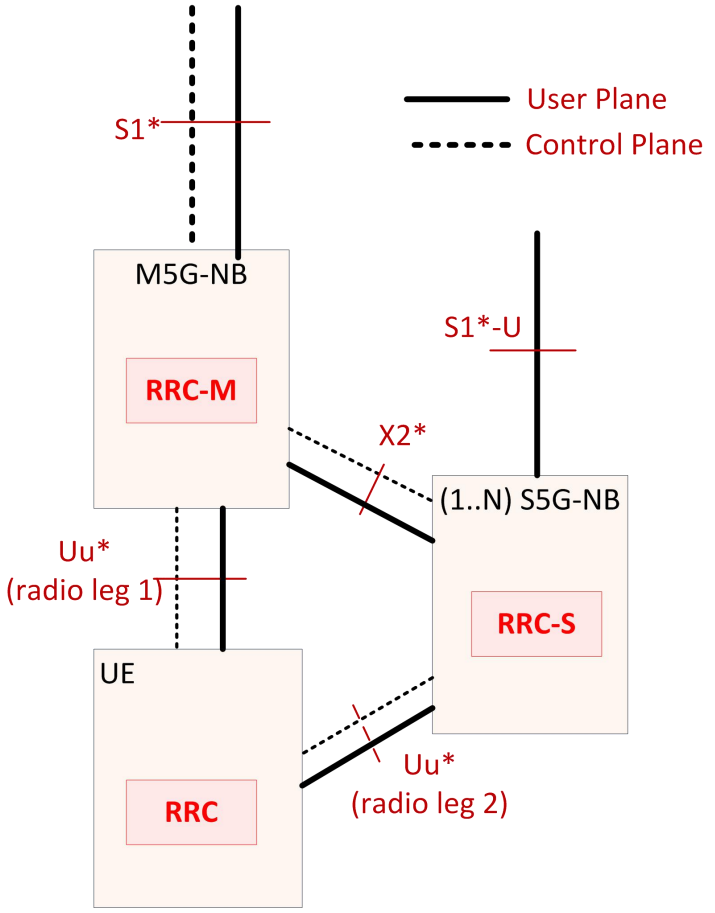
Baseline: Multi-connectivity with signalling reliability
Single RRC



Motivation:
- Multi-connectivity control based on Dual Connectivity principles
- Support for signalling robustness by using multi-connectivity

Principles:
- M5G-NB hosts RRC-Master and is the mobility anchor.
- One or more S5G-NBs provide additional resources (i.e. radio legs) to the UE
- S5G-NB hosts a transparent RRC entity invisible to UE
  - handles radio parameters for S5G-NB Addition Request
  - maintains a local UE context
- Radio leg 1 and Radio leg 2 provide diversity for RRC signalling

Option: RRC-M and RRC-S
Direct message transfer from RRC-S



Motivation:
- Avoid signaling to M5G-NB and CN for path switch
- Fast reconfiguration of S5G-NB radio legs (e.g., intra-S5G-NB HO)
- Termination of measurements at S5G-NB

Role of RRC-M:
- All RRC services and functions (including UE state assignment)
- Coordination with RRC-S

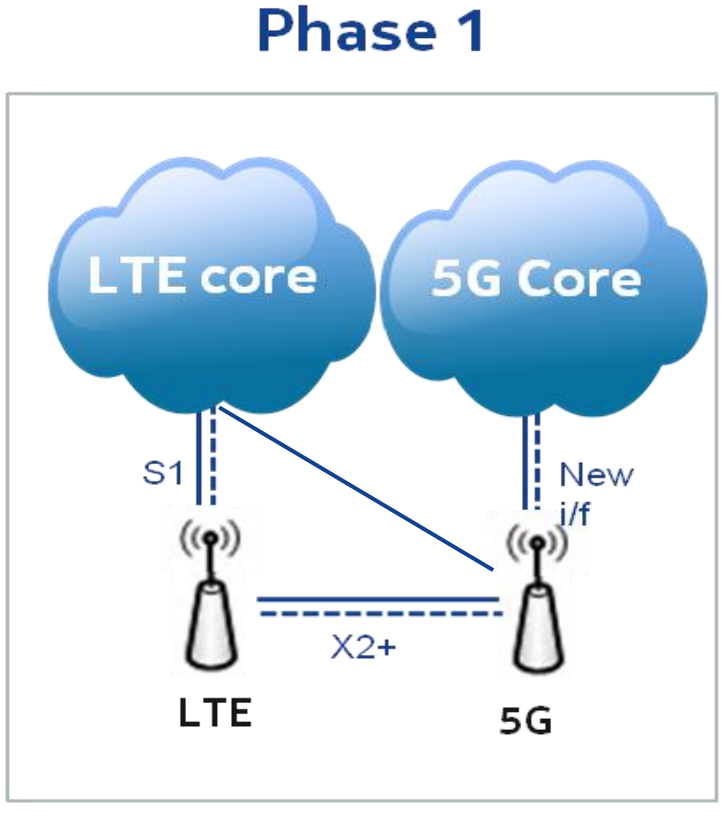
Role of RRC-S:
- Configuration S5G-NB radio legs and resources (pre-agreed with RRC-M)
- Termination of SRF
- Measurement processing for S5G-NB
- Intra-S5G-NB Hand-Over control

Notes:
- One or more RRC-S can be controlled by a single RRC-M
- UE maintains a single RRC state which combines RRC-M and RRC-S connections

LTE-5G and Multi-RAT integration

Phase I: LTE as Anchor + 5G Standalone

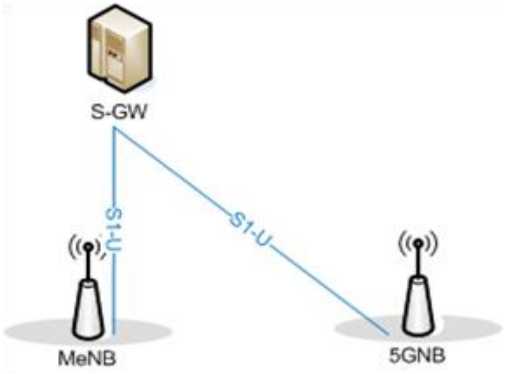
LTE-5G interworking

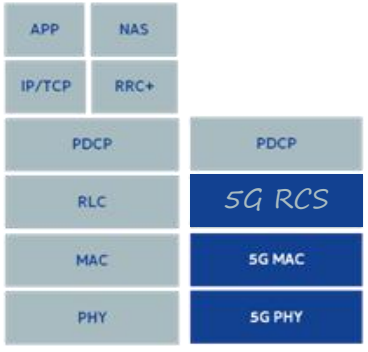


Phase 1: Early 5G radio

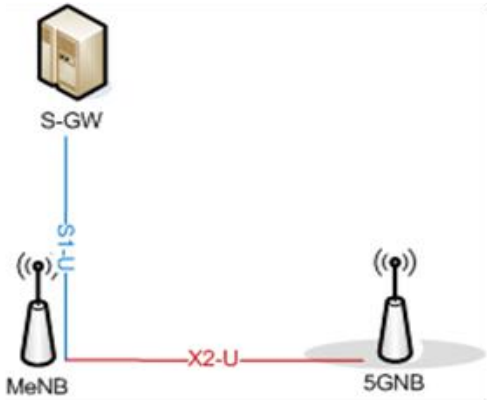
- Early5G radio optimized for eMBB
- 5G & LTE dual **connectivity**
- 5G in standalone mode
- Dynamic QoS framework
- Mobility framework
- Network slicing (Core)
- Flexible scalability of CP/UP functions.
- Minimum feature set to ensure forward compatibility for cMTC and mMTC usecases

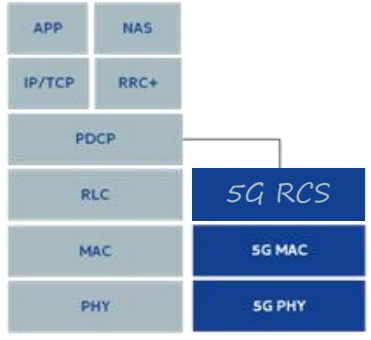
Phase 1 Interworking based on LTE dual connectivity options





5G UE: Option 1a





5G UE: Option 3c

LTE-anchored solution for phase 1 is not required to support new 5G functions for e.g.

- Idle mode
- 5G mobility
- Paging

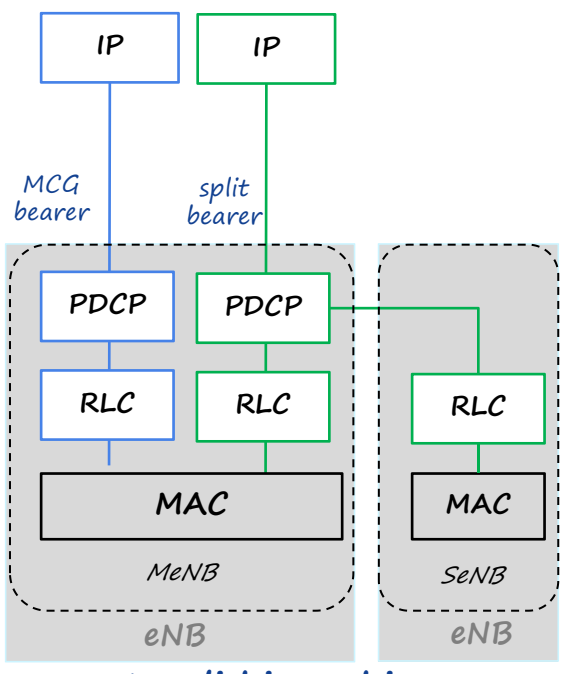
Those functions are LTE based.

PHY, MAC and 5G RCS shall support main 5G features and shall be forward compatible for phase 2 (support of MTC, IoT, V2X, D2D ...).

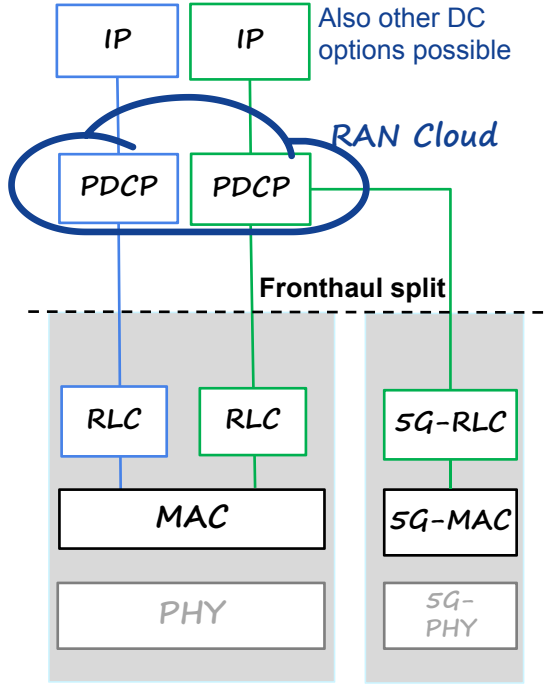
X2-U extensions for 5G are likely → X2+

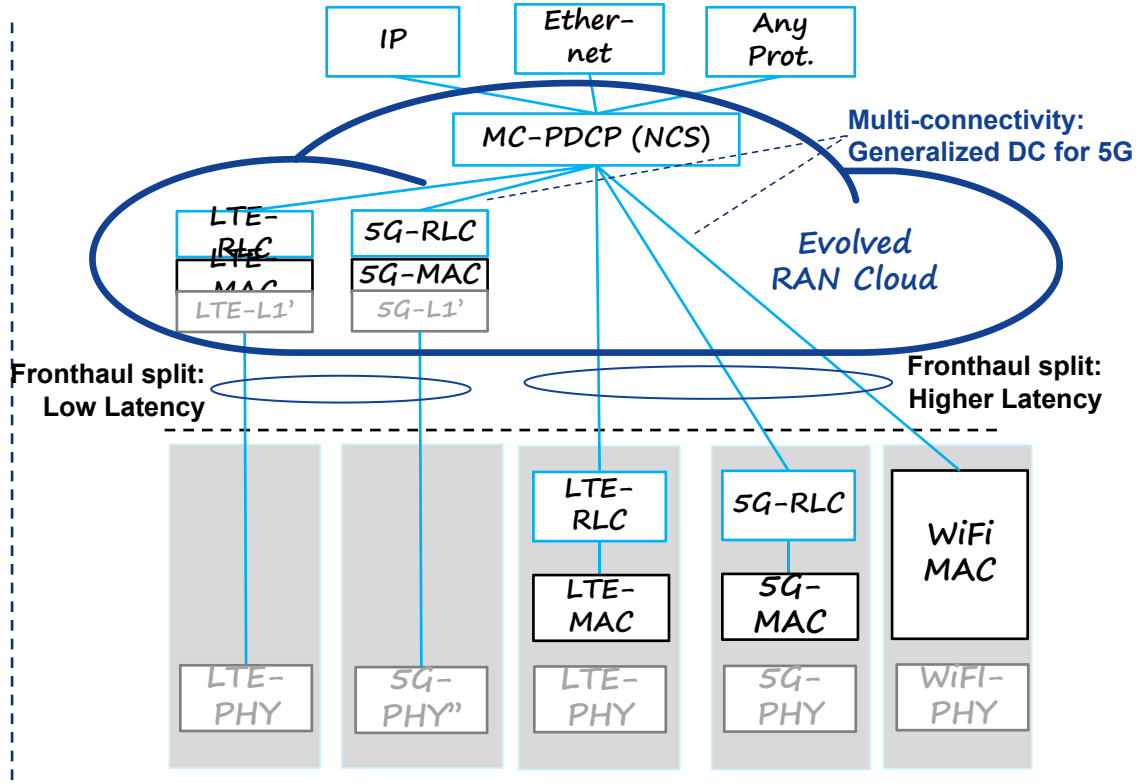
From PDCP to NCS

Evolution of LTE Dual Connectivity



Monolithic architecture



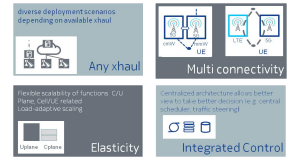


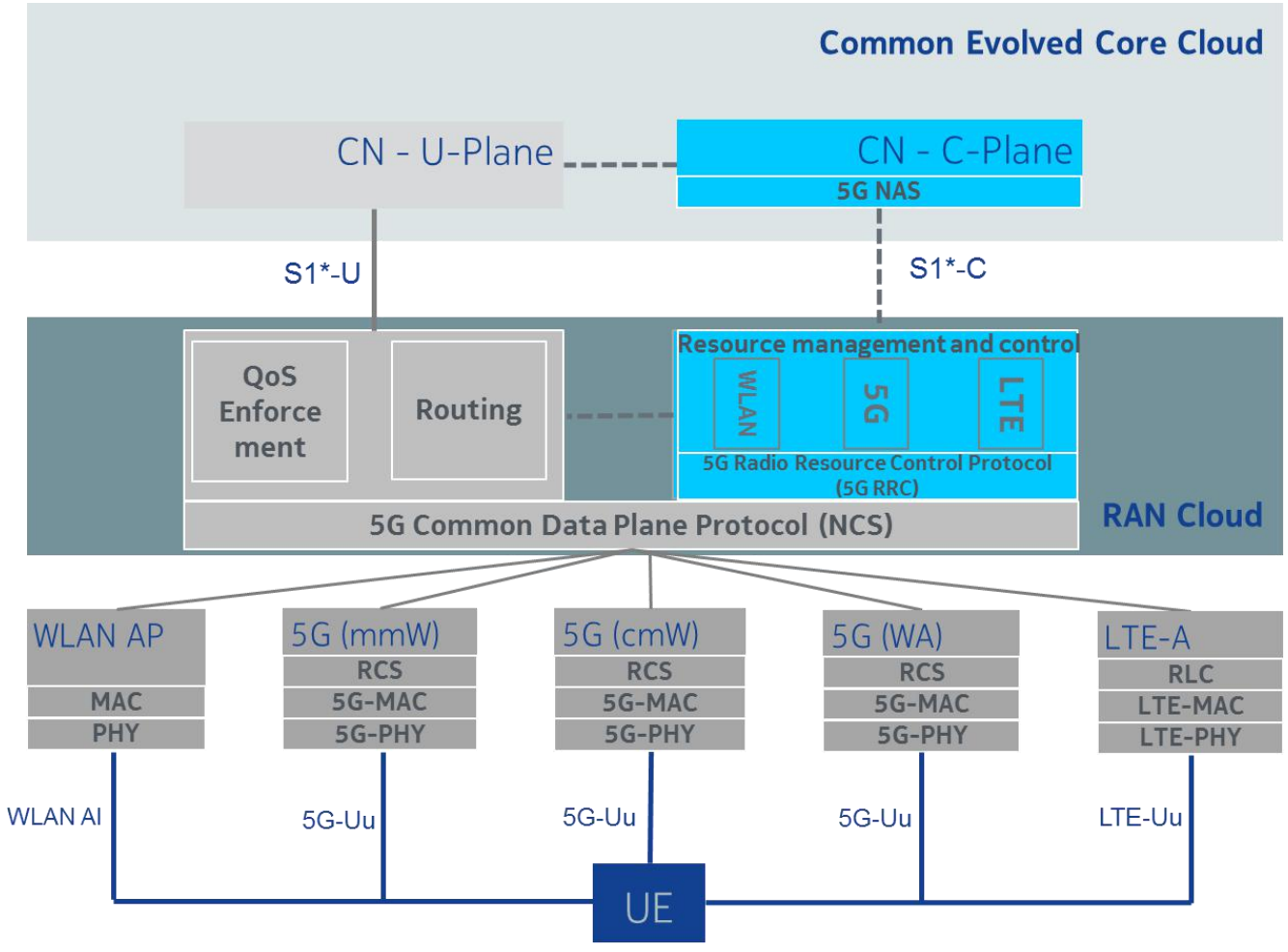
Baseline – LTE Rel. 12 Dual Connectivity

LTE radio cloud with 5G
Scalable and Future proof
Works for high-latency FH

Cloud-based 5G architecture
Generalized Dual Connectivity (Multi-Connectivity)
Same protocol for any RAT: NCS
Scalability in evolved RAN cloud
Coexists with both low-latency and high-latency FH

Phase II: 5G as anchor
Full integration of 5G, LTE, WiFi





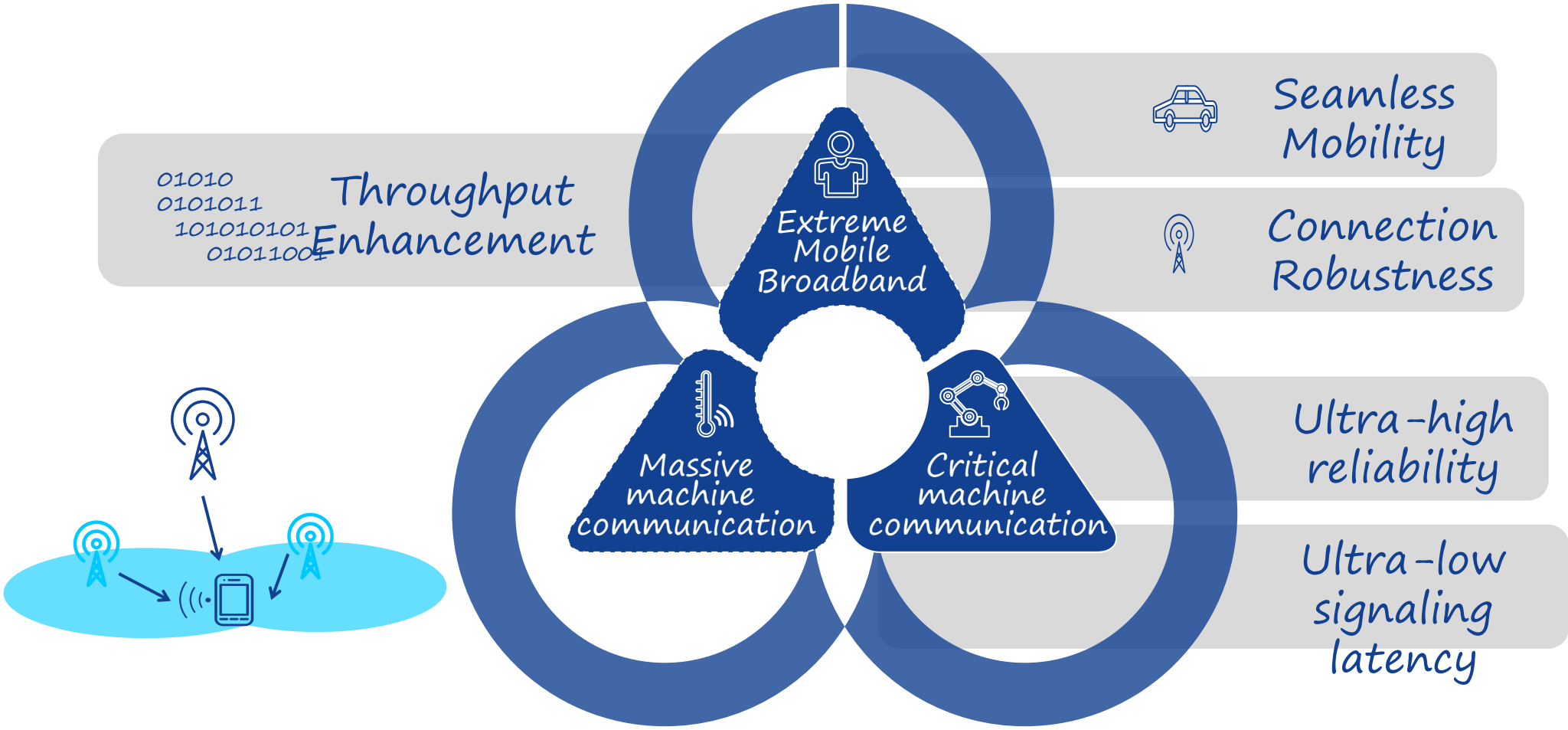
Hybrid macro LTE eNB Protocol stack



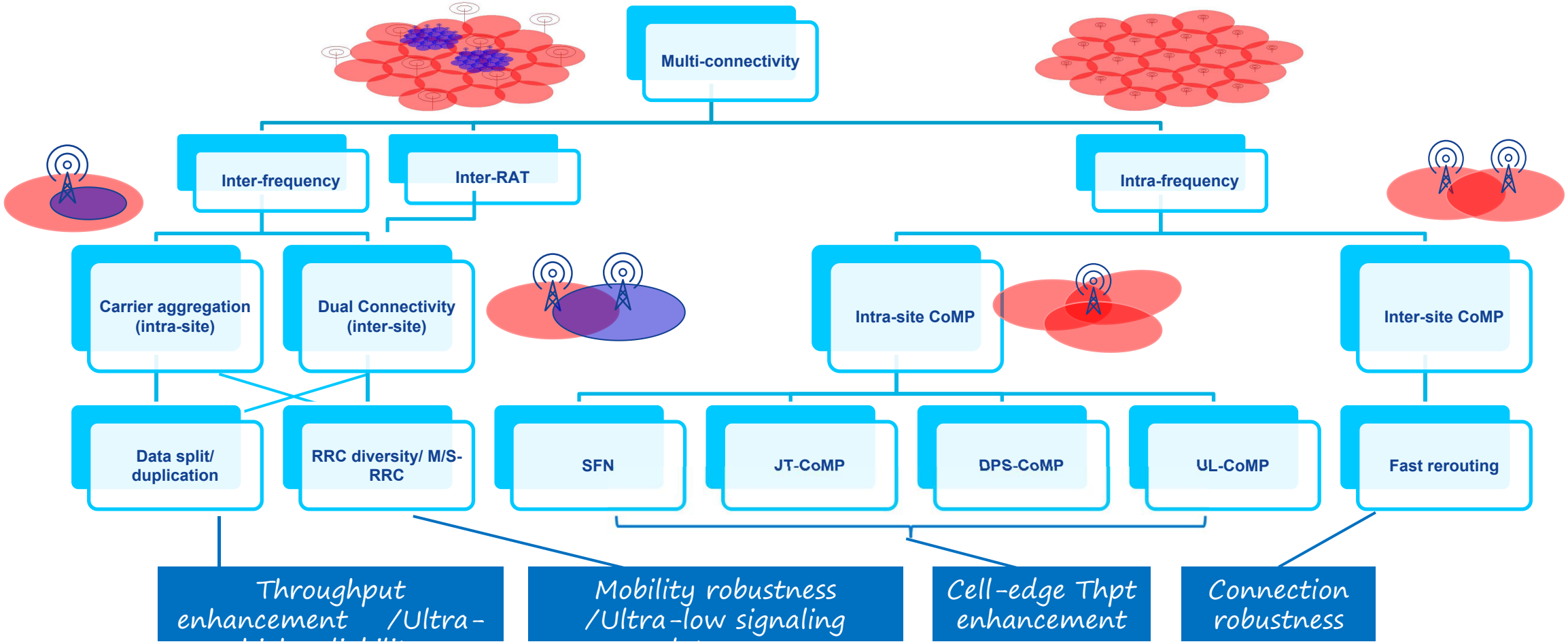
5G UE 4G UE

Intra-5G Multi-Connectivity Options

5G Multi-connectivity – One of the keys to the programmable world

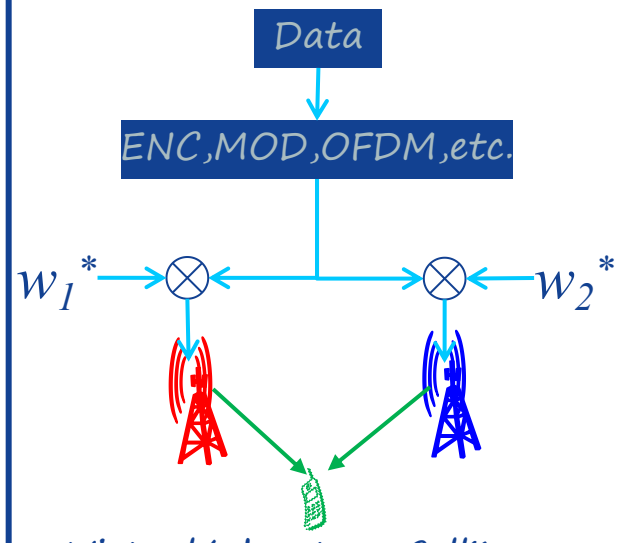


Multi-connectivity Variants for 5G



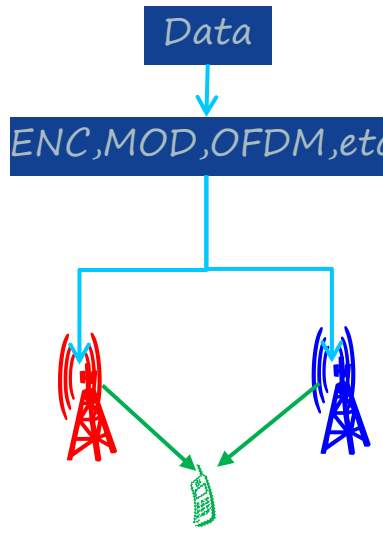
Multi-Connectivity Options – intra-frequency

Joint Transmission



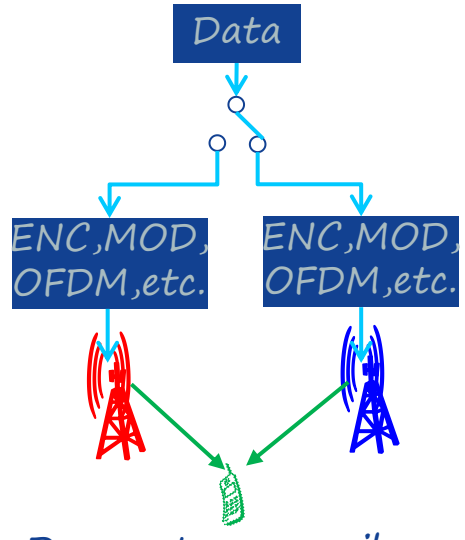
- „Virtual/phantom Cell“
- Coherent FB and calibration
- ~Single cell with distr. antennas
- Joint MAC needed

SFN



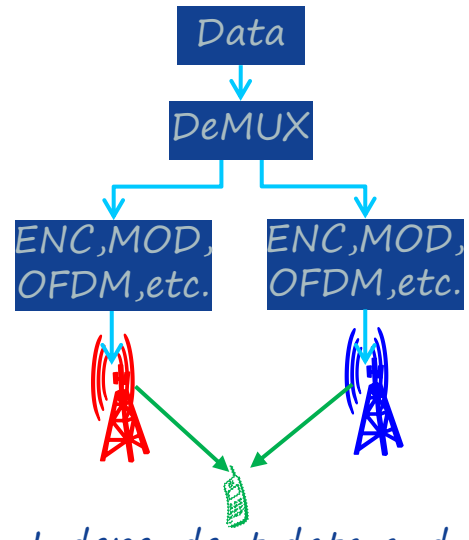
- Classical SFN gain
- Joint MAC needed (Same PRBs must be scheduled!)

Fast Cell Select



- Does not necessarily require joint MAC
- Upper Layer (L2-U) would be slower but still fast...

Multi-link



- Independent data and independent scheduling
- Higher Layer option (lower layer included as MIMO in JT / SFN)

Requires joint layer 2 (L2-L) (= “joint MAC scheduler”)

Possible without joint layer 2 (L2-L)

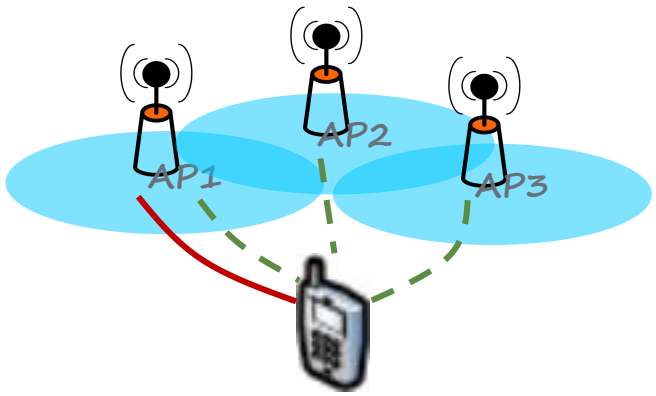
Evolution from LTE dual-connectivity to 5G multi-connectivity

LTE	**Intra-site CoMP** Applied to user-plane To improve cell-edge perf.	**Intra-RI CA** Data split assumed Applied to user-plane For throughput enhancement	**Dual connectivity** Data split assumed Applied to user-plane For throughput enhancement
5G	**Intra-site & inter-site CoMP** Fast rerouting supported Applied also to control-plane To improve mobility robustness	**Intra-RI & Inter-RI CA** Data duplication supported Applied to also control-plane To achieve ultra-high reliability	**Extended DC** Enhanced DC options Data duplication supported RRC dual connectivity defined For connection robustness, ultra-high reliability, ultra-low signaling latency

UP or CP Multi-Connectivity?

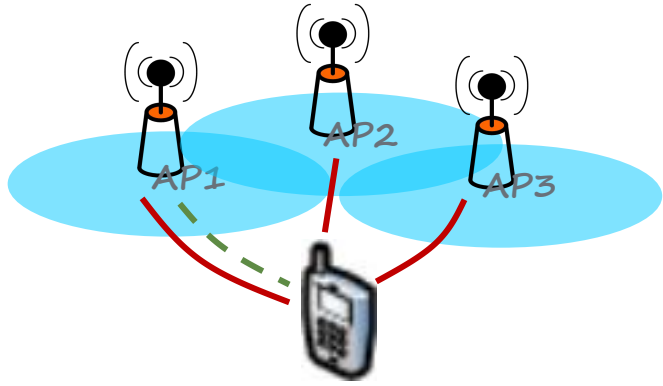
CP
UP

#1.UP-Multi-connectivity



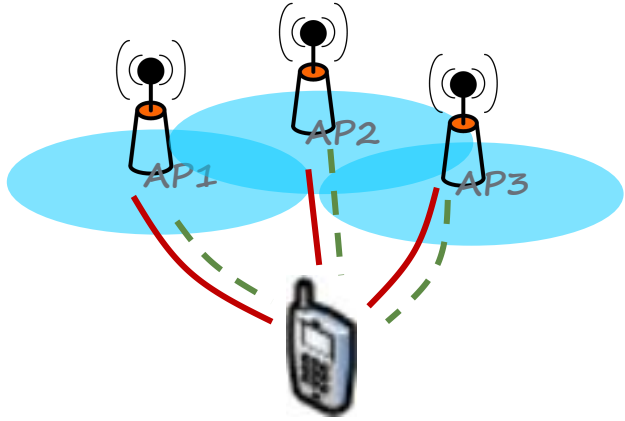
- Aggregating radio resources in more than one eNB for user plan data Tx.
- Control plane as well as mobility still in one cell only.
- Adopted in LTE E.g. CoMP, cell aggregation, CA, Dual connectivity

#2. CP multi-connectivity



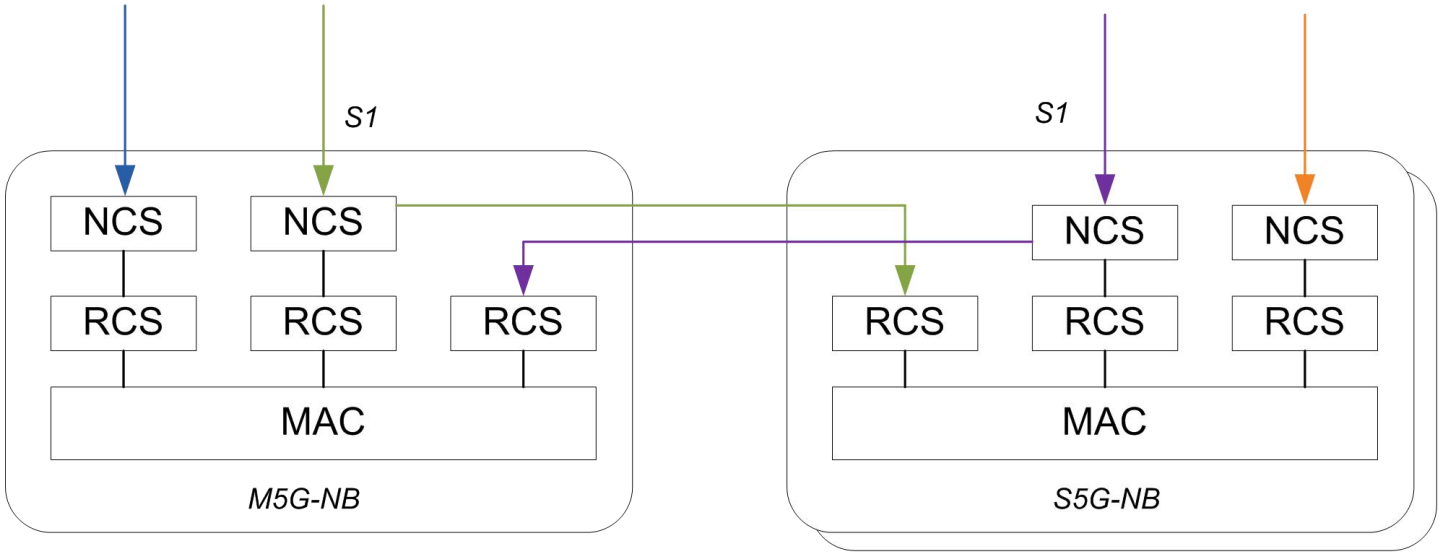
- Handover related RRC signalling could additionally be transmitted from or to a potential target cell e.g. RRC diversity, or multiple RRC
- User plane still in one cell only.
- Not used in LTE yet.

#3. Full



- Both CP and UP connected to multiple cells.
- E.g. Inter-node radio resource aggregation + RRC diversity
- Not used in LTE yet.

Multi-Connectivity
Inter-5G-NB



Protocol architecture

M5G-NB hosts M-RRC.

S5G-NB hosts RCS in case of split radio flow, NCS otherwise.

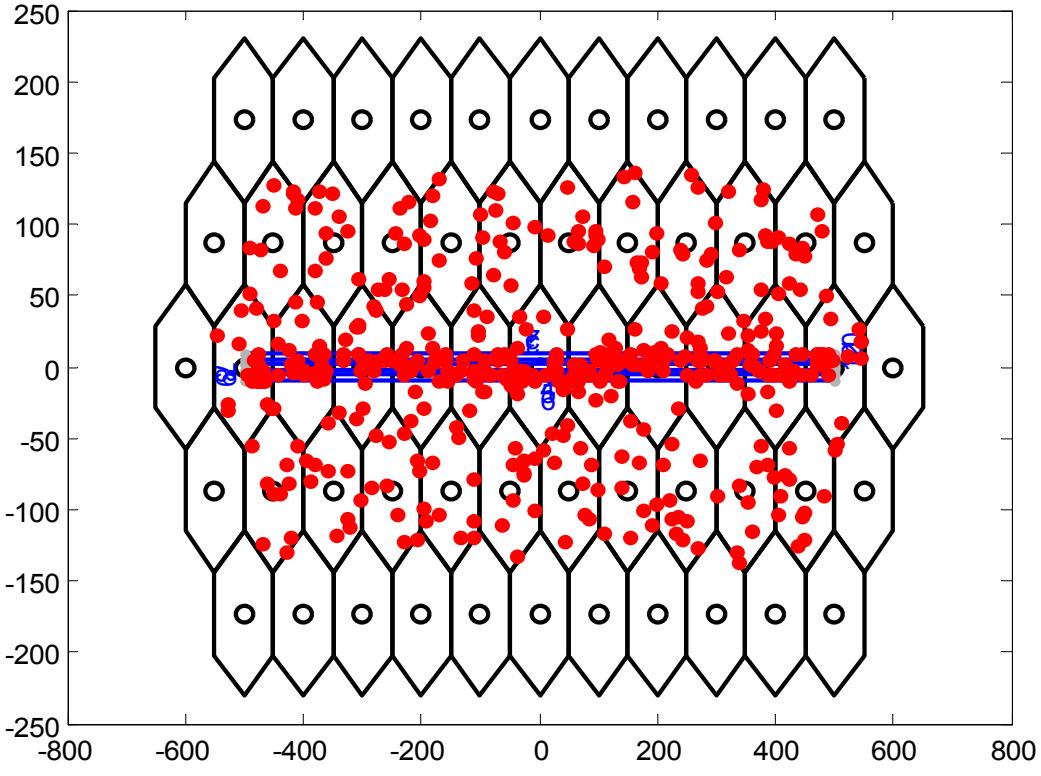
Split Radio Flow can be located at M5G-NB or S5G-NB:

- **S1-U** combined with split radio flow (i.e., in LTE-DC terms, options 1a and 3c combined)
- *FFS* whether split radio flow from S5G-NB to other S5G-NB should be possible.

Mobility robustness through multi-connectivity

Simulation of RLF

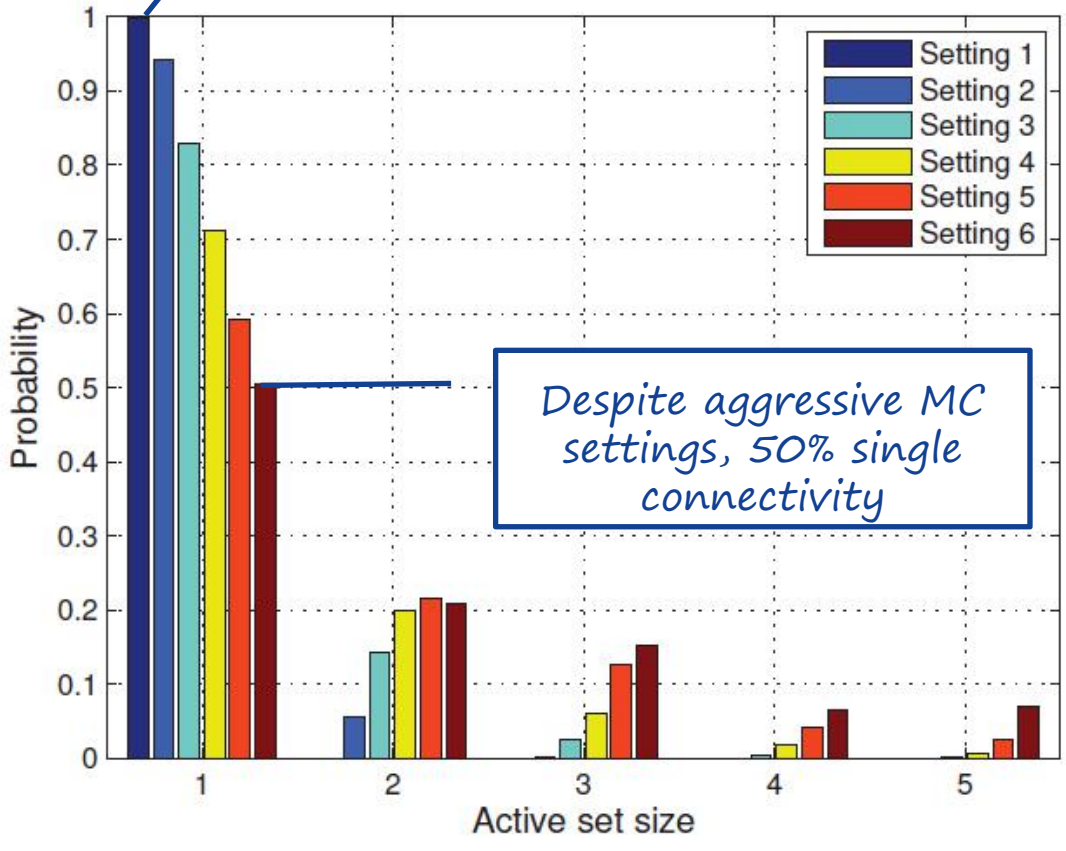
Scenario:



Users

Street

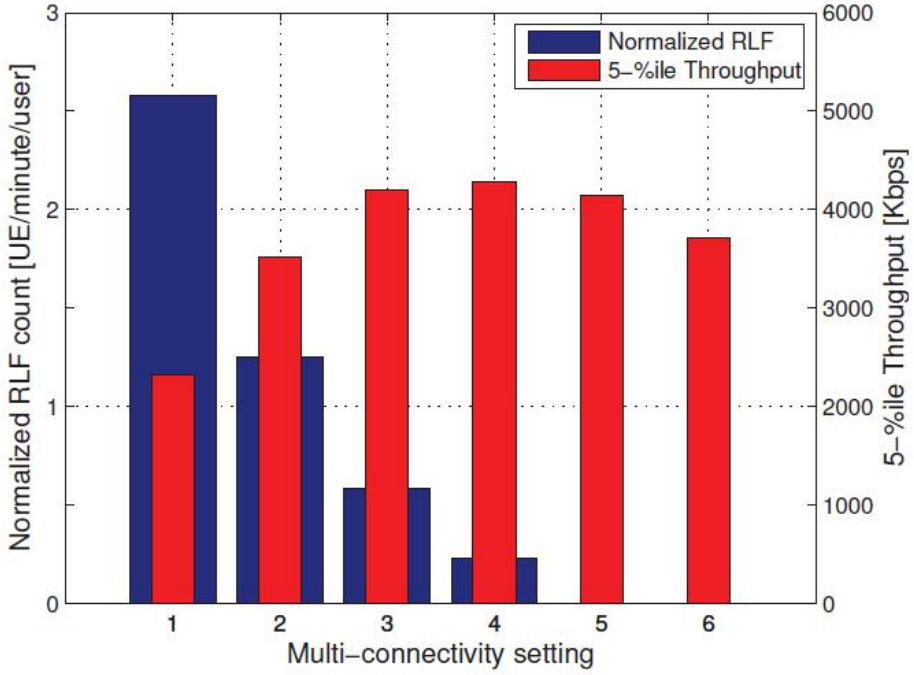
Corresponds to single connectivity

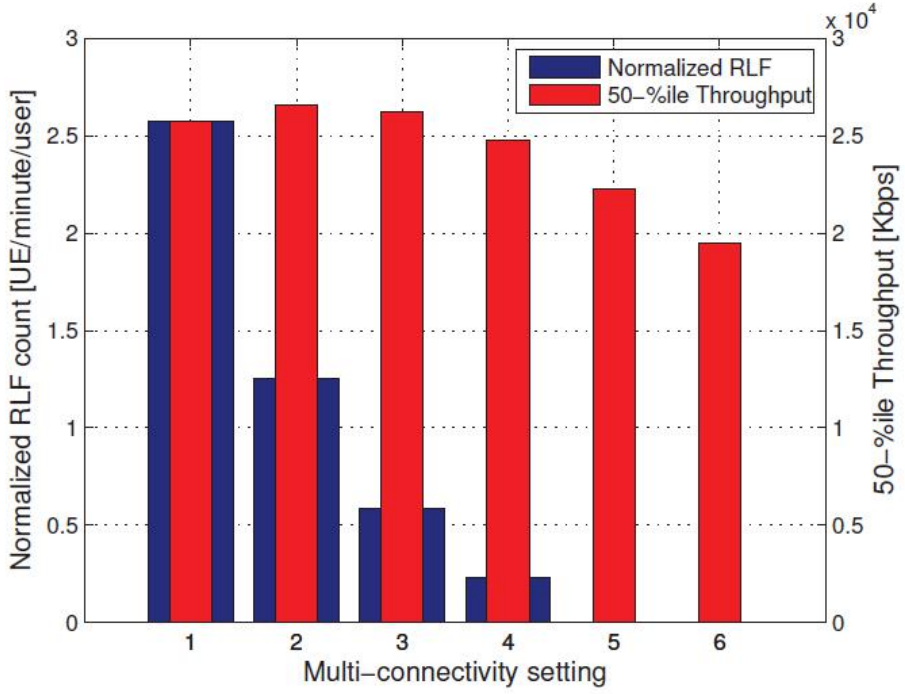


Despite aggressive MC settings, 50% single connectivity

Mobility robustness through multi-connectivity

Radio Link Failure vs Throughput

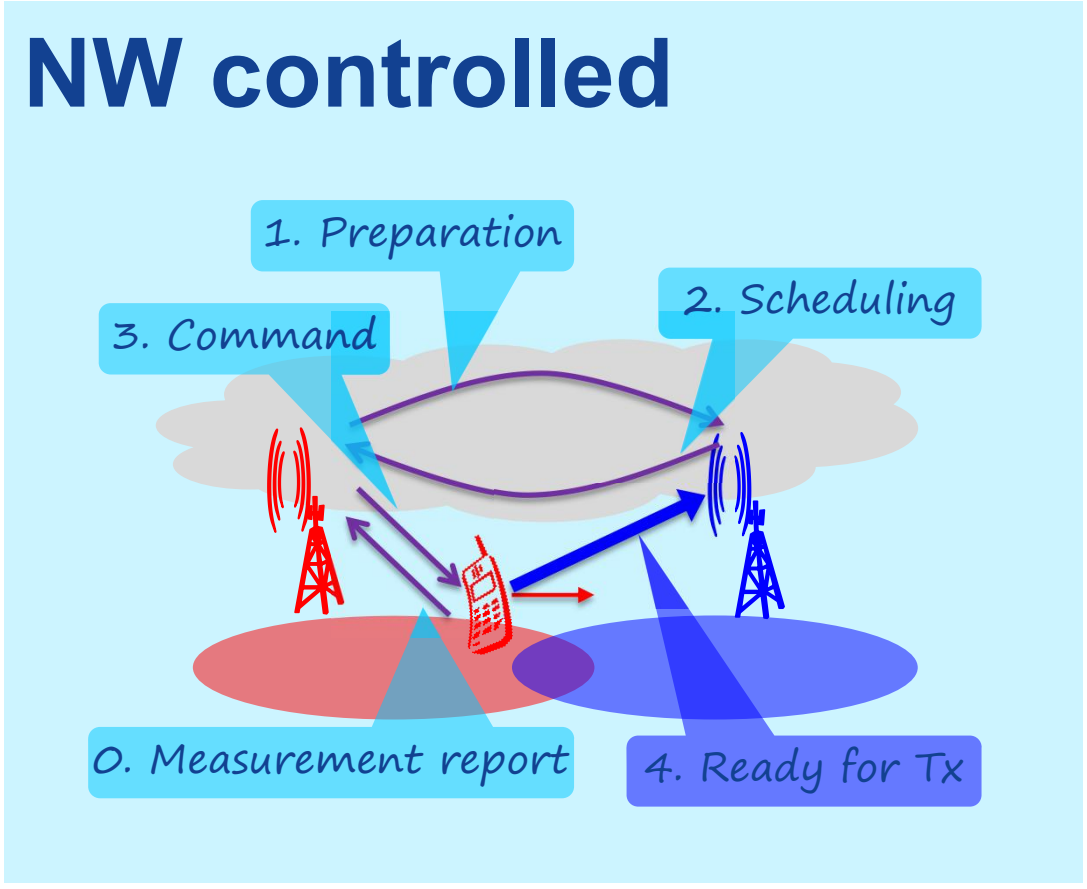


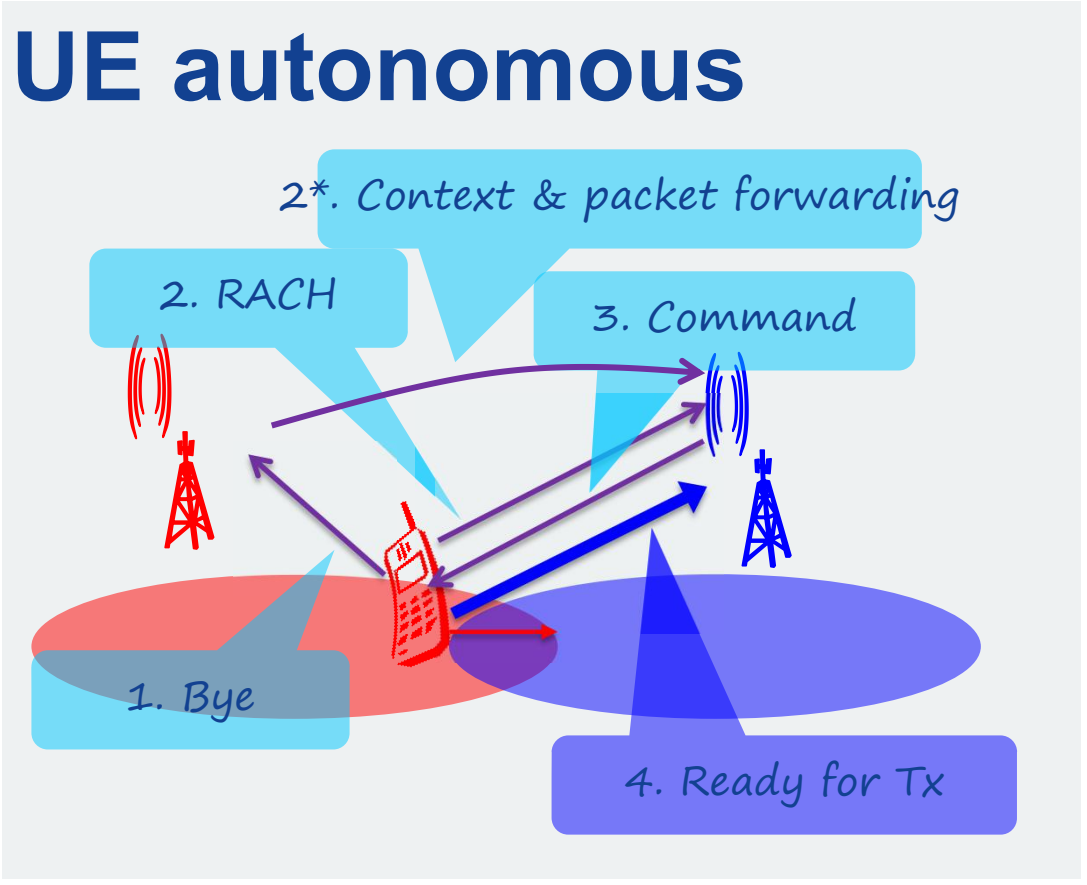


- With MC, 5-% Throughput is improved because of Combined scheduling and low mobility problems.
- As expected, the throughput at relatively higher Active Size configuration decline because resources from more cells are blocked by users of very high active set size.
- Mobility problems are improved with MC

Mobility management
-selected aspects-

Existing Handover Mechanisms: Two mainstreams





Note: authentication information and packet forwarding is not shown in this figure and needed for both

Measurement time: Can it be shortened?

What to measure?

- Slow channel changes (path loss)

What to filter?

- Fast fading, noise …

"Filter" fast fading (Rayleigh fading)

→ Averaging along sliding window

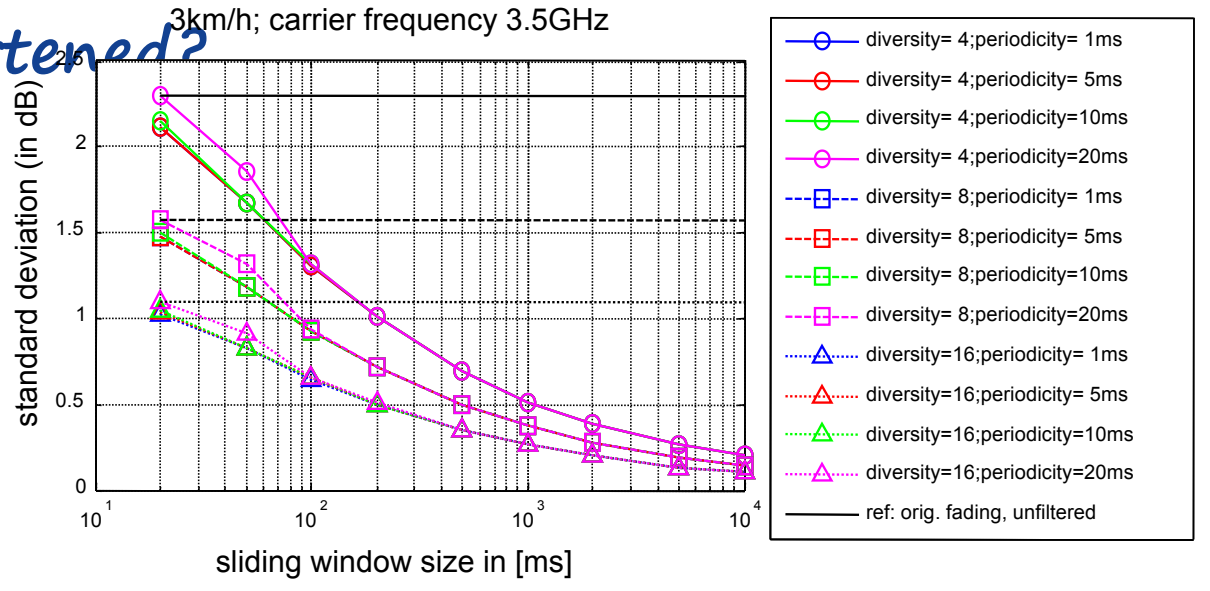
Diversity order:

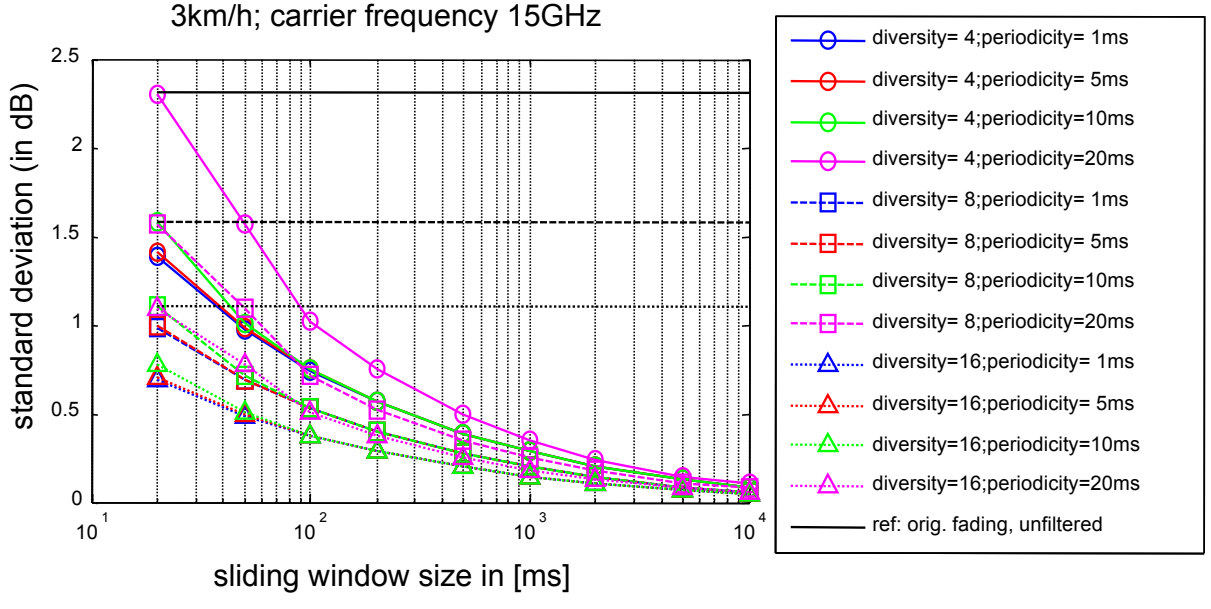
= independent channel realizations (due to MIMO, coherence bandwidth)

Big bandwidth

- Does not improve things: short delay spread → huge coherence bandwidth

→ Only limited reduction of measurement time (10s of ms)





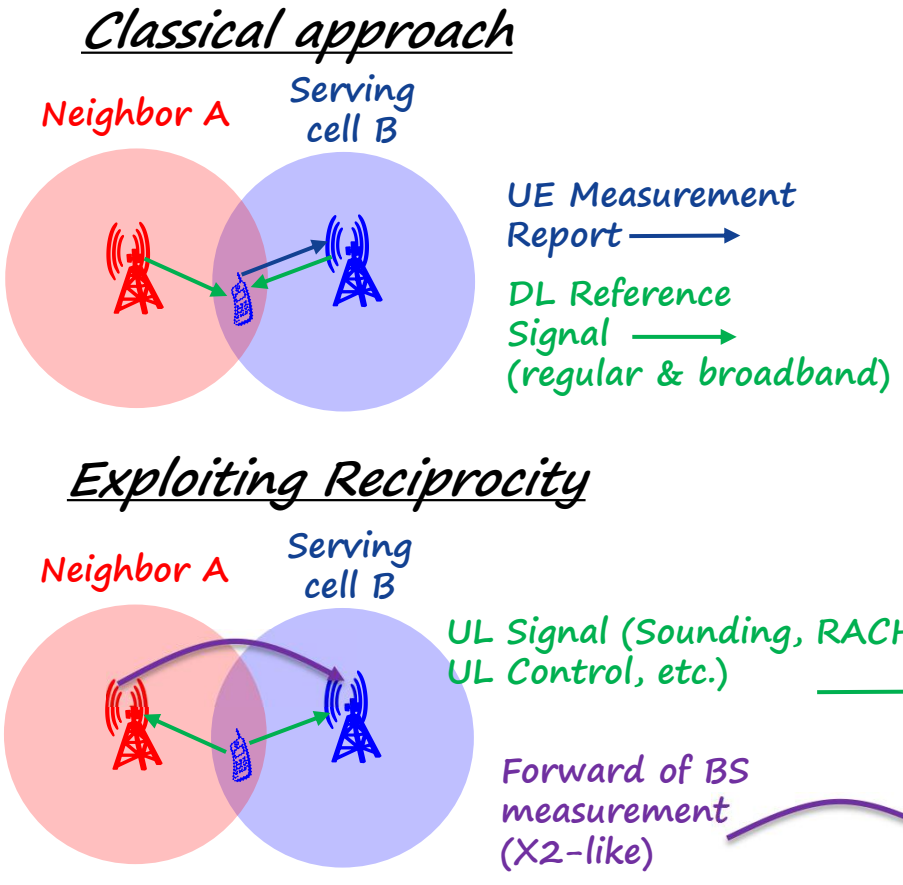
Can channel reciprocity be exploited for measurement?

Uplink can only be measured, if the terminal is

- Transmitting (at all)
- Regularly transmits a broadband signal (just as DL reference signals)
- With well-known power

Conclusions:

- Can NOT rely solely on measurement obtained via reciprocity
  - BS can measure foreign UEs only if "prepared", unexpected cells will be missed and only if BS is also in UL state
- Can use reciprocity to improve/assist mobility in some situations as best effort feature
  - Costs are not significant (if sync is sufficient), at least not in the intra-cloud case



Findings – How to decrease HO latency?

To decrease **HO interruption time**:

1. **RACH-less handover**
2. **Avoiding system information reading**

Note: Since X2 latency varies, the schedule the Target sends in advance through Source (inter-site HO) has to be conservative, or a longer than needed (for the data) resource reservation has to be made

NW Controlled

To decrease **HO interruption time**:

1. **Faster system information reading**
   - *e.g. broadcast neighbor cell BSIs at the Source)*
   - *Source could send an [ACK + Target BSI + schedule] as a response to the Bye message*
2. **1-step RACH**

UE autonomous

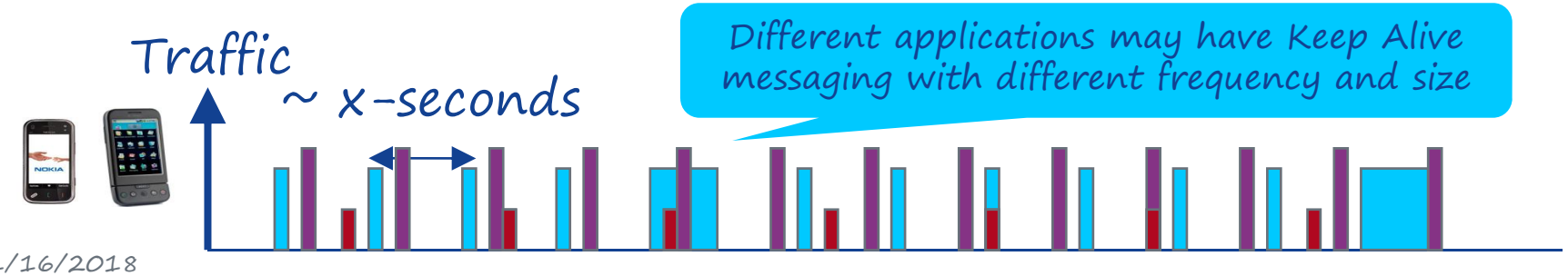
To decrease **HO reaction time**:

1. **Shorter measurement time**:
   - Robustness against failure is dominated by UE measurements.
2. **Minimizing backhaul latency**
   - *If path switch could be initiated earlier, it would save time (it was commented that, in case of HO failure, recovery may be complicated)*

NW & UE

Smart Phone Problem

- Always on Applications:
  - need to send and receive small packets frequently to keep IP connectivity open. This is called **"heart beat"** or **"keep alive"**
  - The typical frequency of "keep alive" is once per minute, or once every few minutes, and the amount of data is very small (<<1 Kbyte)
  - Keep alive messages are typically generated by the application without any option for the network to control them
- "Smart" problem with "always-on" applications:
  - Multiple applications run on top of each other with independent heartbeats
  - This causes frequent IDLE to CONNECTED mode state transition, at the cost of increased C-plane (26 in LTE) signalling in the network, higher power consumption and insufficient radio resource usage.



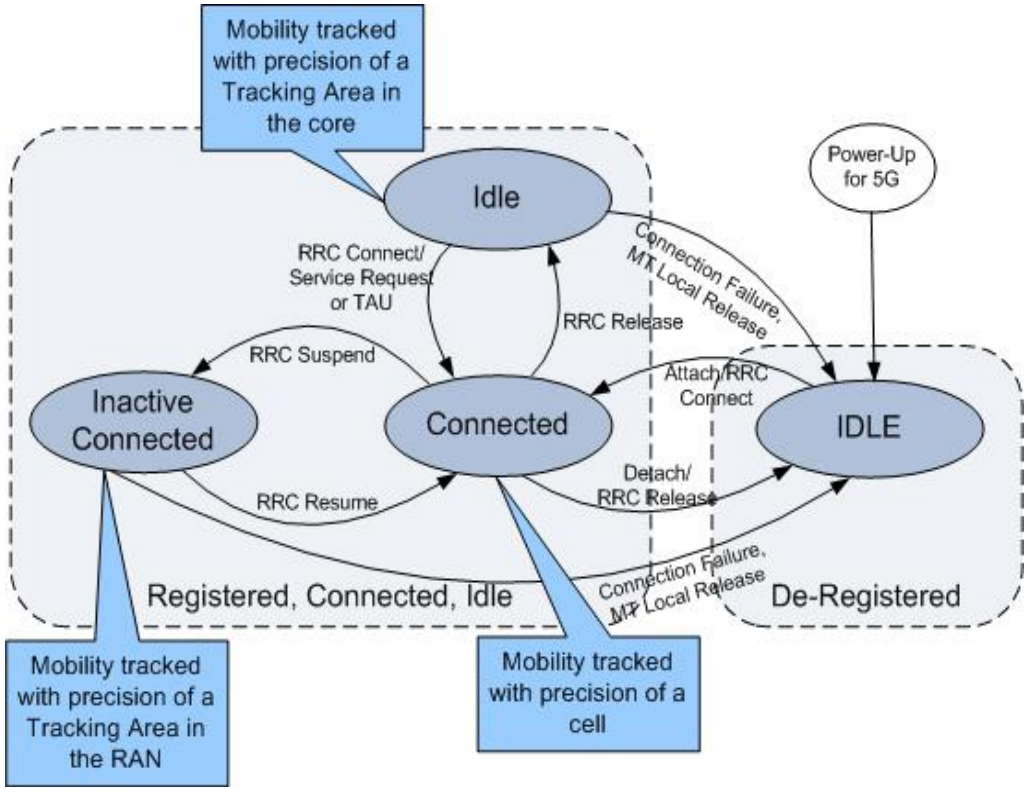
instant messaging, push email, location services, weather widgets etc.

Characteristics of a new RRC Connected Inactive state in 5G

- Potential characteristics of the new RRC Connected Inactive state:
  - **Configurable** based on the requested service by the device
  - Supports **inactive user plane** while Mobility Management (EMM) and Connection Management (ECM) are active
  - **UE based mobility** with UE measurements and cell reselection procedure with configuration from network
  - When configured for the **fast system access** the device will monitor paging or the dedicated signalling.
  - **RAN stores the UE context** during the RRC Connected Inactive state
  - RRC Connected Inactive to RRC Connected **state transition with a light-weight signaling procedure**

5G RRC state management for Connected Inactive

5G State Transition



LTE States

UE status	Off	Attaching	Idle/ Registered	Connecting to EPC	Active
EMM	Deregistered		Registered		
ECM	Idle				Connected
RRC	Idle	Connected	Idle	Connected	

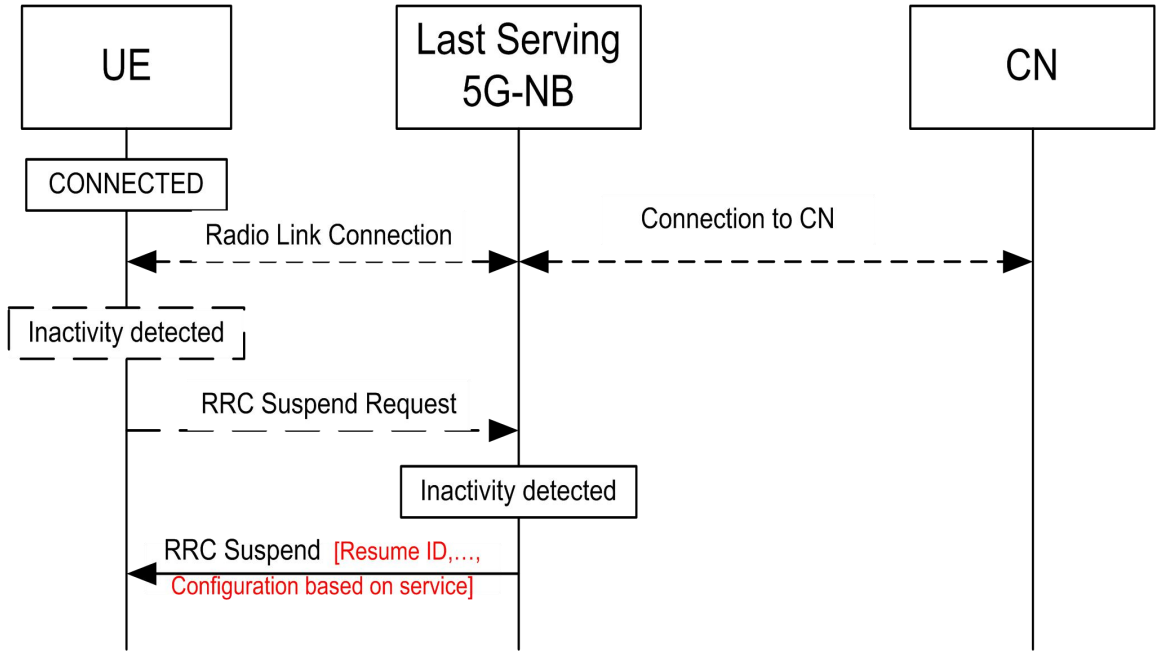
5G States

UE status	Off	Attaching	Idle/ Registered	Connected/ Inactive	Connected/ Active
EMM	Deregistered		Registered		
ECM	Connected				
RRC	-	Connected	Idle	Intermediate	Connected
Mobility control	UE based		UE based, NW assisted		NW based

EPS Mobility Management (EMM)
EPS Connection Management (ECM)
Evolved Packet System (EPS)
Evolved Packet Core (EPC)

Configurability of a new RRC Connected Inactive state in 5G

- Potential parametrization and configurability aspects of the new RRC Connected Inactive state:
  - **Mobility/location tracking management configuration:** RAN based mobility and location tracking, single/multiple cell-level tracking
  - **Measurement configuration:** Measurement configuration for cell reselection, camping, etc taking into account the existence of multiple air interface variants
  - **Camping configuration:** Single/multiple-RAT camping, capacity based camping, etc
  - **State transition/system access configuration:** State transition and RACH access optimizations
  - **Synchronization configuration:** DL and/or UL synchronization



A service could be characterized based on its requirements, e.g., on mobility, security & privacy, reliability, bandwidth, latency, battery life, etc.

Questions



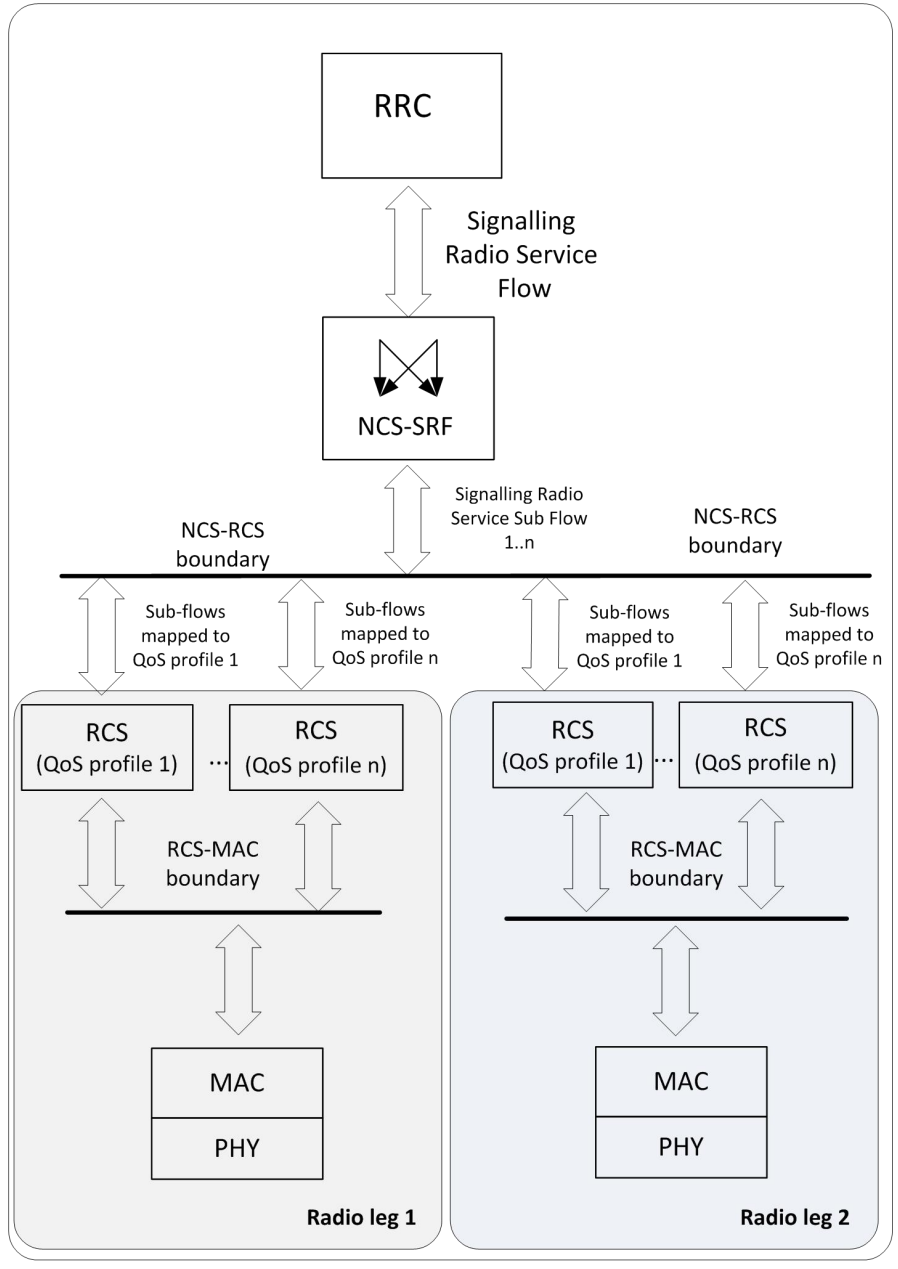
Backup

Signalling Radio service Flows

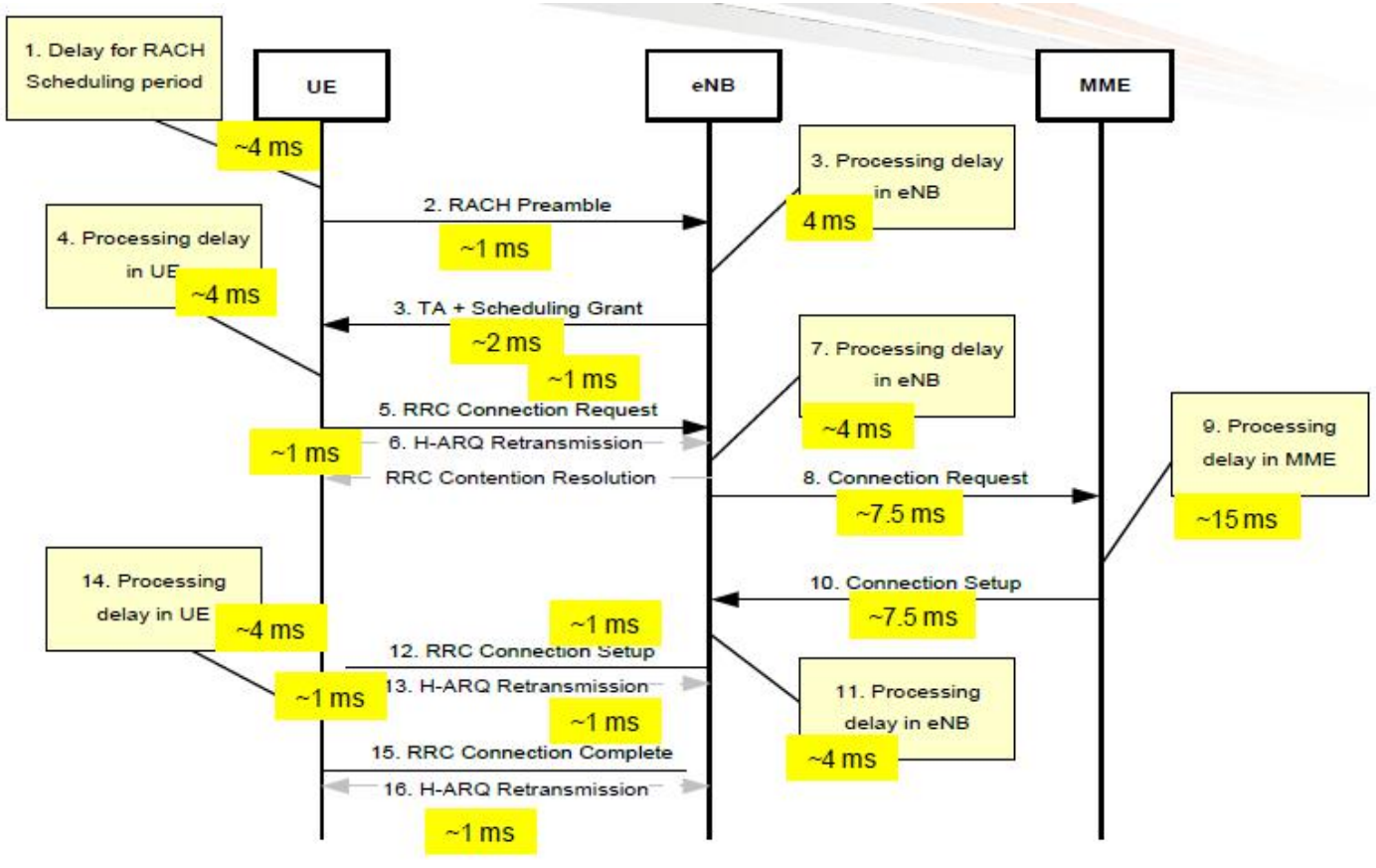
- SRF is an aggregation of signalling sub-flows for transmission of RRC and NAS messages.
- An SRF is established during the process of RRC connection establishment and (possibly) reconfigured later on by the network.
- An SRF may contain one or more Signalling Radio service Sub-flows. One example is:
  - RRC messages using the CCCH logical channel;
  - RRC messages with high priority DCCH logical channel;
  - RRC messages with low priority DCCH logical channel;

NOTE:

- NAS messages transferred via Signalling Radio service Sub-flows are also contained in RRC messages.
- RRC protocol is used for configuring the NCS and adding or removing Signalling Radio service Sub-flows.
- Signalling Radio service Sub-flows could have independent or same RCS (configuration choice). E.g. when some SSubF has to have independent properties of the others.



RRC States ~ Transition from Idle to Connected



RRC_IDLE to RRC_CONNECTED state transition, messages required for the UE to transfer/receive user plane data.

Some factors affecting C-plane latency during connection setup

- Contention-based or-less system access
- Continuous or discontinuous monitoring of paging messages or scheduling grants.
- Processing delays at RAN and Core
- Protocol architecture

Definitions and Abbreviations

Basic definitions

5G-NB: 5G-NodeB is a logical entity in 5G-RAN responsible for radio transmission / reception in one or more cells to/from the User Equipment.

5G-RAN: 5G-RAN is the Radio Access Network to support very high data rates, ultra-low-latency and massive MTC.

5G-Uu: The radio interface between UE and 5G-RAN.

Cell Group (CG): A group of serving cells associated with a 5G-NB and configured by the network.

NOTE: A CG may only include cells that are associated to the same 5G-NB, and those cells are synchronized at the 5G-NB level similar as for carrier aggregation. In distributed 5G-NB configuration, a cell group can be associated to a 5G-RAP.

Logical Channel: A logical channel is an information stream dedicated to the transfer of a specific type of information over the radio interface. Logical Channels are provided on top of the MAC layer.

Network convergence sublayer (NCS): A sublayer of OSI layer 2 which responsible for routing and adaptation of higher layer traffic to different radio interfaces. It corresponds to PDCP in LTE.

Network convergence sublayer – SRF (NCS-SRF): The NCS instantiated for the purpose of transporting the SRF.

Radio convergence sublayer (RCS): A sublayer of OSI layer 2 in 5G-RAN which is responsible for buffering, link reliability, segmentation and concatenation. It corresponds to RLC in LTE.

Radio Flow (RFL): The service provided by NCS for transfer of data between User Equipment and 5G-UTRAN.

Radio Sub Flow: A radio sub flow is part of a radio flow between UE and 5G-RAN, which is mapped to a specific RCS Queue. In case of multi-connectivity, a radio sub flow can be mapped to RCS Queues in different radio legs.

RCS Queue: An RCS queue transports information with a specific QoS and priority. RCS Queues also include the buffer of information in 5G-RAP. An RCS queue is mapped to a Logical Channel.

Signaling Radio Flow (SRF): Signalling Radio Flow is defined as a Radio Flow that are used only for the transmission of RRC and NAS messages. An SRF is an aggregation of one or more Signalling Radio Sub-Flows.

Transport Channel: The channels offered by the physical layer to Layer 2 for data transport between peer L1 entities are denoted as Transport Channels. Different types of transport channels are defined by how and with which characteristics data is transferred on the physical layer, e.g. whether using dedicated or common physical channels.

Transport network layer: Entirety of functions in a network for forwarding packets between physical interfaces of network elements. In an overlay architecture, frames/packets of a service flows are encapsulated and tunneled over the transport network layer between service network layer functions.

Fronthaul and Sidehaul related

Distributed 5G-NB: A 5G-NB which supports Fs interface and optionally, Si interface.

5G-RAP: 5G Radio Access Point is a logical node in 5G-RAN which terminates the Fs interface from MNC and the 5G-Uu interface. It partially implements the 5G-RAN protocol stack. A 5G-RAP is part of a 5G-NB.

Fs: The logical interface for Fronthaul split between MNC and 5G-RAP.

MNC: Multi-Node-Controller is a logical entity in 5G-NB which terminates the Fs interface towards the 5G-RAP and the S1 interface towards CN. It hosts the UE and cell specific control-plane and user plane functions of a 5G-NB, including specific RRM and call processing functions.

MNC-C: The MNC-C comprises UE and cell specific C-Plane functions of an MNC. The MNC-C manages and controls 5G-RAPs.

MNC-U: The MNC-U comprise user-plane functions of the MNC.

Si: The logical interface between an MNC-C and an MNC-U.

Multi-connectivity definitions

M5G-NB: In Inter-5G-NB, the 5G-NB which instantiates M-RRC per UE.

MCG: The CG of the M5G-NB.

Multi-Connectivity: Mode of operation of a UE in RRC_CONNECTED, configured with resources in two or more cells. Two cases can be differentiated:

- Inter-5G-NB: Multi-Connectivity involving one M5G-NB, and one or more S5G-NB.
- Intra-5G-NB: Multi-Connectivity involving cells of the same logical 5G-NB.

  NOTE: Inter- and Intra-5G-NB multi-connectivity can be configured simultaneously.

Multi-Connectivity-Link: The data link between the multi-connectivity anchor and the Ue. The link comprises Fs or X2* (when applicable) and 5G-Uu (radio) interfaces.

NCS-Multi-Connectivity: Mode of operation where NCS controls traffic handling for multi-connectivity.

NCS-Multi-Connectivity-Link: The Multi-Connectivity Link if NCS is the multi-connectivity anchor sublayer.

Radio-Leg: In case of Multi-Connectivity, a cell or group of cells configured with resources for a UE. It is expected that the cells of a Radio Leg belong to the same Radio Interface (mmW, cmW, WA).

  NOTE: A radio leg is the 5G-Uu (radio) part of a multi-connectivity link.

S5G-NB: In Inter-5G-NB Multi-Connectivity, a 5G-NB with SCGs.

SCG: A group of serving cells of the S5G-NB, controlled or coordinated by M-RRC.

Split radio flow: A RFL which is split into two or more radio subflows.

Split radio sub flow: A RSFL whose radio resources are located in more than one radio legs.

References

References

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